

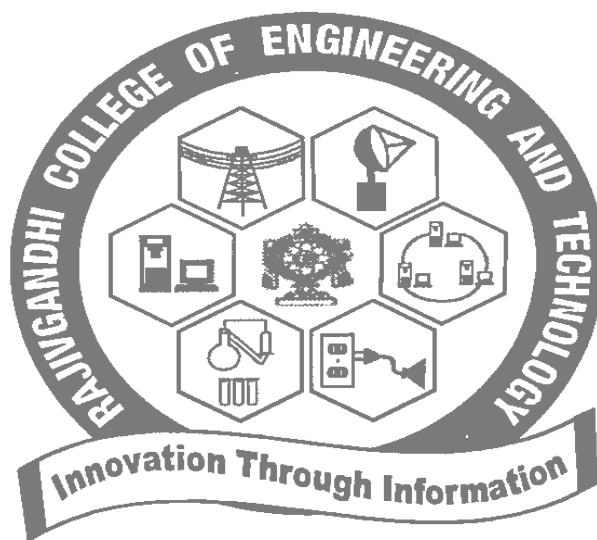
**RAJIV GANDHI COLLEGE OF ENGINEERING
AND TECHNOLOGY**

KIRUMAMPAKKAM

PONDICHERRY- 607 402

PHYSICS LAB MANUAL

(FIRST YEAR B.TECH)



DEPARTMENT OF PHYSICS

PHYSICS LABORATORY

CONTENTS

Sl. No.	DATE	NAME OF THE EXPERIMENT	MARK	Viva Mark	SINGNATURE
1.		Spectrometer – Dispersive Power of a Prism			
2.		Newton's Rings			
3.		Vibrational Magnetometer			
4.		Deflection Magnetometer			
5.		Thermal conductivity of a bad conductor – Lee's disc method			
6.		Spectrometer – Determination of wavelength of mercury spectrum using Grating			
7.		Air Wedge			
8.		Polarimeter			
9.		Laser – Determination of wavelength of a given laser source using diffraction Grating			
10.		Thermal conductivity of Rubber by Radial flow method			

MEASURING INSTRUMENTS

(A) MEASUREMENT OF LENGTH

For the measurement of length usually metre scales are used with an accuracy upto millimeters. To measure length say $1/100^{\text{th}}$ of a centimeter (or) $1/100^{\text{th}}$ of a millimeter, it is not possible to measure accurately using metre scale. Hence the following instruments are used for more accuracy.

1. Vernier Calipers (Accuracy upto $1/100^{\text{th}}$ of cm)
2. Screw gauge (Accuracy upto $1/100^{\text{th}}$ of mm)

A1. VERNIER CALIPERS

Design of the Instrument

It consists of two scales viz, main scale and vernier scale. The main scale is a long steel bar graduated in inches on the top and in centimeters on the bottom. 10 main scale divisions are equal to 1cm. The vernier scale has 10 divisions which are equal to 9 main scale divisions. There are two metal jaws A & B. A is fixed and B is movable which can be fixed at any position using the screw (s) as shown in fig. A-1.

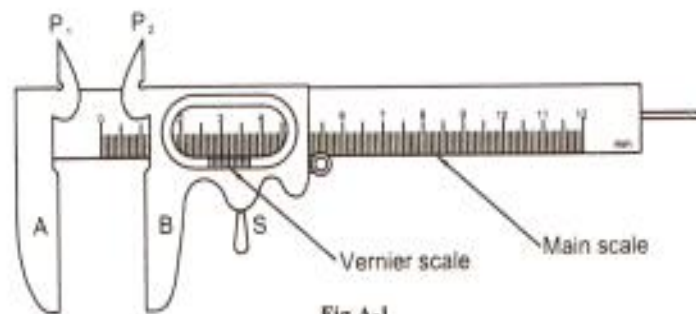


Fig A-1

The projections P_1 and P_2 of the jaws in the upward direction is used to measure the internal diameter of the calorimeter, cylinder, etc.

Least Count

Least Count of vernier calipers is defined as the smallest length that can be accurately measured and is equal to the difference between the main scale division (MSD) and the vernier scale division (VSD).

Derivation

$$LC = 1 \text{ MSD} - \text{VSD}$$

$$10 \text{ MSD} = 1 \text{ cm}$$

$$10 \text{ MSD} = \frac{1}{10} \text{ cm}$$

$$\text{But } 10 \text{ VSD} = 9 \text{ MSD}$$

$$\therefore 1 \text{ VSD} = \frac{9}{10} \text{ MSD (or) } \frac{9}{10} \left(\frac{1}{10} \right)$$

$$= \frac{9}{100} \text{ cm}$$

$$LC = \left(\frac{1}{10} - \frac{9}{100} \right) \text{ cm}$$

$$= \frac{1}{100} \text{ cm} = 0.01 \text{ cm}$$

$LC = 0.01 \text{ cm.}$

No Zero Error

To find the zero error of the vernier calipers, the two jaws A and B are brought in contact with each other. If the zero of the vernier scale coincides with the zero of the main scale then there is no zero error (fig. A-2). Hence there is no need to apply zero correction.

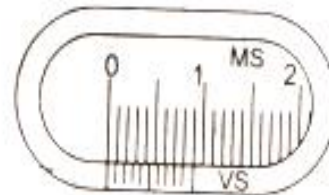


Fig A-2

Positive Zero Error

If the zero of the vernier scale lies on the right side of the zero of the main scale, then the instrument has an error called positive zero error (fig A-3). This error should be subtracted from the final reading. Thus, the zero correction is negative.

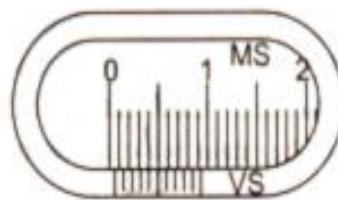


Fig A-3

Example

If the 4th vernier division coincides with any of the main scale division, then

$$\begin{aligned}\text{zero error} &= \text{Vernier Scale coincidence} \times \text{Least count} \\ &= \text{VSC} \times \text{LC} \\ &= 4 \times 0.01 = 0.04 \text{ cm}\end{aligned}$$

$$\therefore \text{Zero Correction} = -0.04 \text{ cm}$$

Negative Zero Error

If the zero of the vernier lies on the left side of the zero of main scale, then the instrument is said to have an error called negative zero error. (fig A-4). This error should be added to the final reading. Thus, the zero correction is positive.

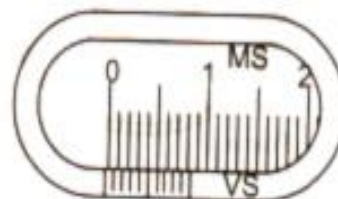


Fig A-4

Example

If the 6th division of the vernier coincides with any of the main scale division, then

$$\begin{aligned}\text{Zero error} &= -(\text{number of vernier scale divisions} - \\ &\quad \text{vernier scale coincidence}) \times \text{least count} \\ &= -(10 - 6) \times 0.01 \text{ cm} \\ &= -4 \times 0.01 = -0.04 \text{ cm}\end{aligned}$$

$$\text{Zero Correction} = 0.04 \text{ cm}$$

To find the length of the specimen

The specimen whose length (or) outer diameter is to be determined is held between the two jaws. The position of zero of the vernier on the main scale gives the main scale reading (MSR). The vernier division which coincides with any one of the main scale division gives the vernier scale coincidence (VSC), then

$$\text{Observed reading (OR)} = \text{MSR} + (\text{VSC} \times \text{LC}) \text{ in cms}$$
$$\text{Actual (or) correct Reading} = \text{OR} \pm \text{Zero Correction in cms}$$

B. SCREW GAUGE

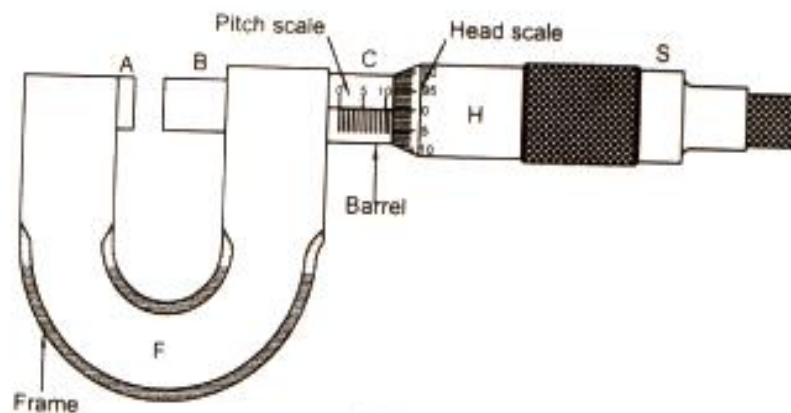


Fig B-1

Design of the Instrument

It consists of two scales namely pitch scale and head scale. Pitch scale is a millimeter scale engraved on the cylinder (C), which is rigidly attached with the frame (f). The head scale carries 100 equal divisions. The specimen can be held inbetween the two edges A (fixed) and B (movable) as shown in fig.B-1.

Derivation of Least Count

$$\text{LC} = \frac{\text{Pitch}}{\text{No. of head scale divisions}}$$

$$\text{Pitch} = \frac{\text{Distance moved on the pitch scale}}{\text{No. of head scale divisions}}$$

To find pitch the head is given say, two rotations and the distance moved by the head on the pitch scale is noted

$$\begin{aligned}\text{Pitch} &= \frac{2 \text{ mm}}{2} \\ &= 1 \text{ mm}\end{aligned}$$

Number of head scale divisions = 100

$$\text{LC} = \frac{1}{100} \text{ mm}$$

$$\text{LC} = 0.01 \text{ mm}$$

To find zero error

To find zero error the stud A and the screw tip B are kept in contact. If the zero of the head scale coincides with zero of the pitch scale on the reference line, the instrument is said to have no zero error (fig. B-2).

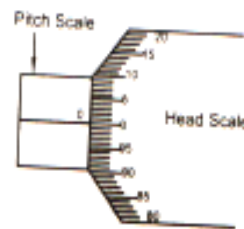


Fig B-2

Positive zero error

If the zero of the head scale lies below the reference line of the pitch scale, the zero error is positive and the correction is negative. (fig B-3).

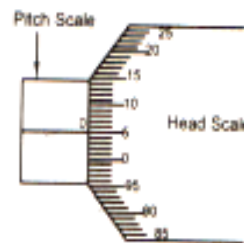


Fig B-3

Example

If the 5th division of the head scale coincides with the reference line of the pitch scale then

$$\text{Zero error} = \text{head scale coincidence} \times \text{Least count}$$

$$= 5 \times 0.01 \text{ mm}$$

$$= 0.05 \text{ mm}$$

$$\text{Zero correction} = - 0.05 \text{ mm}$$

Negative zero error

If the zero of the head scale lies above the reference line of the pitch scale (fig B-4) the zero error is negative and the correction is positive.

Example

If the 95th division of the head scale coincides with reference line of the pitch scale then,

$$\text{Zero error} = -(100 - \text{head scale coincidence}) \times \text{least count}$$

$$= -(100 - 95) \times 0.01 \text{ mm}$$

$$= -0.05 \text{ mm}$$

$$\text{Zero correction} = 0.05 \text{ mm}$$

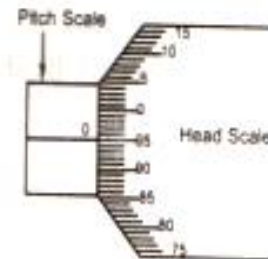


Fig B-4

To find the diameter of the given wire

The wire is gripped gently between the faces A and B. The number of pitch scale divisions just in front of the head scale gives the pitch scale reading. The division on the head scale that coincides with the reference line gives the headscale coincidence. The readings are noted and are tabulated.

Example

Suppose if the PSR = 5 mm

$$\text{H.S.C.} = 35 \text{ div,}$$

$$\text{HSR} = \text{HSC} \times \text{L.C.} \Rightarrow 35 \times 0.01 = 0.35 \text{ mm}$$

$$\text{Observed reading} = \text{PSR} + \text{HSR} \Rightarrow 5 + 0.35 = 5.35 \text{ mm}$$

If the zero error is 0.04 mm, then the zero correction is -0.04 mm,

$$\begin{aligned} \therefore \text{Correct reading} &= \text{OR} \pm \text{Z.C} \\ &= (5.35 - 0.04) \text{ mm} \\ &= 5.31 \text{ mm} \end{aligned}$$

C. TRAVELLING MICROSCOPE

Design of the Instrument

Travelling Microscope consists of an ordinary compound microscope which slides along a graduated vertical pillar, attached to a horizontal base resting on the levelling screws. The main scale divisions along with vernier scale divisions are marked on the horizontal base and the vertical pillar as shown in fig.C-1.

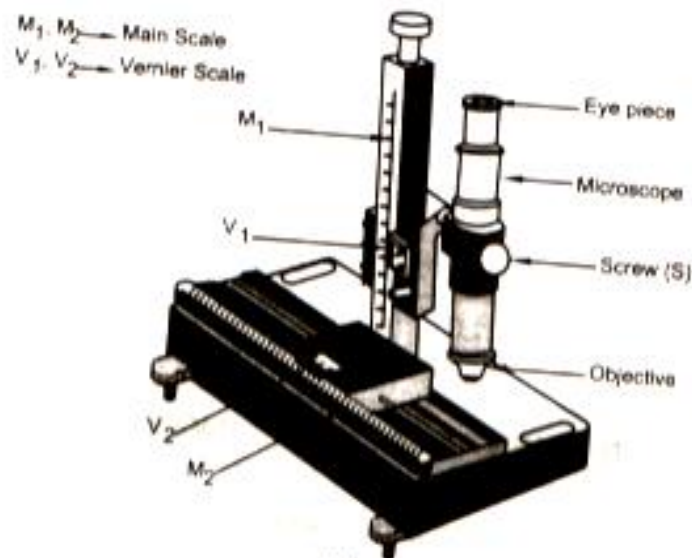


Fig C-1

The microscope can be moved up and down in the vertical pillar and can be moved in, to and fro direction over the horizontal base. Thus the microscope can be moved both in the vertical and horizontal directions. Two fine adjustment screws are provided for the horizontal and vertical movements respectively.

The image of the object can be focused by adjusting the side screw (S) attached to the microscope. The eye piece of the microscope is provided with a cross wire.

The main scale is divided into millimeter and half a millimeter. Therefore the value of one mainscale division (MSD) is 0.5 mm. The Vernier scale of the travelling microscope is divided into 50 divisions which are equivalent to 49 main scale divisions.

Least Count Derivation

$$LC = 1 \text{ MSD} - 1 \text{ VSD}$$

$$20 \text{ MSD} = 1 \text{ cm}$$

$$\therefore 1 \text{ MSD} = \frac{1}{20} \text{ cm}$$

$$\text{Here } 50 \text{ VSD} = 49 \text{ MSD}$$

$$\therefore 1 \text{ VSD} = \frac{49}{50} \text{ MSD (or)} \frac{49}{50} \left(\frac{1}{20} \right) \text{ cm}$$

$$1 \text{ VSD} = \frac{49}{1000} \text{ cm}$$

$$\therefore LC = \left(\frac{1}{20} - \frac{49}{1000} \right) \text{ cm}$$
$$= \frac{1}{1000} \text{ cm}$$

$LC = 0.001 \text{ cm}$

D. MEASUREMENT OF ANGLES

Spectrometer - Design of the instrument

Spectrometer consists of collimator (c) telescope (T), prism table (p) and vernier table as shown fig D-1. The collimator consists of a convex lens fixed at one end and a slit of adjustable width on the other end.

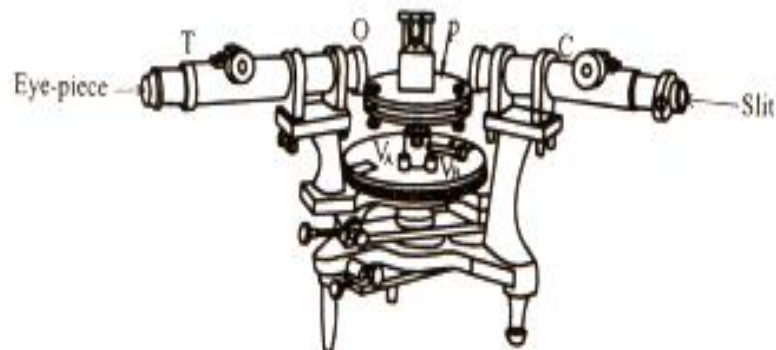


Fig D-1

The telescope consists of an objective(O) at one end and an eye piece fixed with the cross wires on the other end. The prism table consists of two circular disc connected by three levelling screws. The vernier tables has two verniers V_A and V_B each having main scale and vernier scale. Here one main scale division is equal to half a degree. Each vernier scale has 30 divisions, which is equal to 29 main scale divisions.

Initial adjustments

- (i) **Eye Piece adjustment:** The eye piece (E) is adjusted until the cross wires are clearly seen when viewed through the telescope.
- (ii) **Distant object method:** A clear, well defined inverted and diminished image of a distant object is seen by adjusting the telescope.
- (iii) **Slit adjustment:** The slit is made narrow with the help of the screw, provided aside of the slit.
- (iv) **Collimator adjustment:** The telescope is brought on line with the collimator. The slit is illuminated by a source of light. If the image of the slit appears blurred, then the screw of the collimator is adjusted until a clear image is seen when viewed through the telescope. Now the rays of light emerging from the collimator will be rendered parallel.
- (v) **Prism table adjustment:** The spirit level is placed on the prism table, parallel to the line joining any two levelling screws. The air bubble in the spirit level is brought to the centre by adjusting the two screws. It is then placed in a perpendicular direction and the air bubble is brought at the centre by adjusting the third screw. Now the prism table will be horizontal.
- (vi) **Spectrometer base:** The base of the spectrometer is adjusted to the horizontal with the help of the three levelling screws.

Least count

$$LC = 1 \text{ MSD} - 1 \text{ VSD}$$

$$1 \text{ MSD} = \frac{1}{2} \text{ degree}$$

Here $30 \text{ VSD} = 29 \text{ MSD}$

$\therefore 1 \text{ VSD} = \frac{29}{30} \text{ MSD}$

$$1 \text{ VSD} = \frac{29}{30} \times \frac{1}{2} \text{ degree}$$

$$1 \text{ VSD} = \frac{29}{60} \text{ degree}$$

$$\text{LC} = \left(\frac{1}{2} - \frac{29}{60} \right) \text{ degree}$$

$$= \frac{30-29}{60} \text{ degree}$$

$$= \frac{1}{60} \text{ degree}$$

$$\text{Since } 60' = 1^\circ$$

$$1' = \frac{1}{60}^\circ$$

$$\therefore \boxed{\text{LC} = 1'}$$

Measurement

Before doing any experiment using spectrometer, the above mentioned initial adjustments have to be made. While performing the experiment, the mainscale and vernier scale readings are noted from both the verniers in V_A and V_B in degrees and minutes.

$$\therefore \text{Total Reading} = \text{MSR} + (\text{VSC} \times \text{LC}) \text{ degrees}$$

SPECTROMETER – DISPERSIVE POWER OF A PRISM

Expt No:

Date :

AIM:

To determine the dispersive power of a given Prism

APPARATUS REQUIRED:

Spectrometer, Glass Prism, Mercury Vapour Lamp, Reading Lens etc.,

FORMULA:

(i) Refractive index of the prism, $\mu = \frac{\sin\left(\frac{A+D}{2}\right)}{\sin\left(\frac{A}{2}\right)}$ (No unit)

(ii) Dispersive power of the prism, $\omega = \frac{\mu_2 - \mu_1}{\mu - 1}$ (No unit)

(iii) The resolving power of the prism R.P. = $t \left(\frac{d\mu}{d\lambda} \right)$

A-Angle of the prism (deg)

D-Angle of minimum deviation (deg)

μ_1 & μ_2 - Refractive indices of two colours. (no unit)

t – base length of the prism

$d\mu$ change in refractive index of two colours

$d\lambda$ change in wavelength of colours

PROCEDURE:

Preliminary adjustments of the spectrometer are made as given below.

(i) Eyepiece Adjustments:

The spectrometer is turned towards a bright object say a white wall 3 to 4m away and the eyepiece is adjusted so that the crosswire are clearly seen.

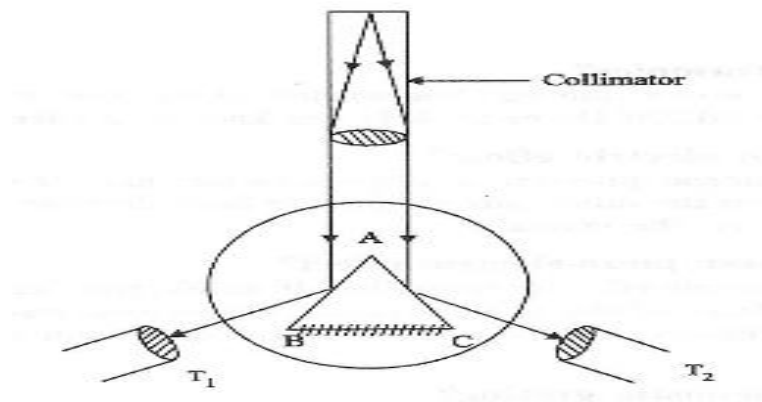
(ii) Telescope Adjustments

The telescope is turned towards a distant object and is adjusted until the distant object is clearly seen. Thus the telescope is set for receiving the parallel rays of the collimator.

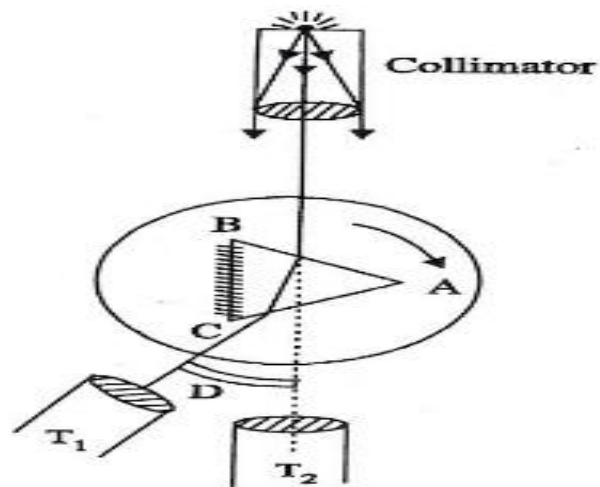
(iii) Collimator Adjustments:

The slit of the collimator is illuminated with a monochromatic source. The telescope is turned to view the image of the slit and the collimator screw is adjusted such that a clear image of the slit is obtained. The slit of the collimator must be adjusted to be vertical and narrow. The initial leveling of the prism is done using a spirit level.

Diagram: To find the Angle of the Prism



To find the angle of Minimum Deviation



Determination of the Angle of the Prism (A):

The given Prism is mounted vertically at the center of the prism table with the refracting edge facing the collimator. Now the parallel rays of light emerging out from the collimator falls almost equally on the two faces of the prism as shown in the figure. The telescope is turned to catch the reflected image from one face of the prism and fixed in that position. The tangential screw is adjusted until the vertical cross-wire coincides with the fixed edge of the image of the slit.

The readings on both the verniers are noted. Similarly the readings corresponding to the reflected image of the slit on the other face are also taken. The difference between the two readings of the same vernier gives twice the angle of the prism. Hence, the angle of the prism A is determined.

Determination of the Angle of Minimum Deviation (D):

The prism table is rotated so that the beam of light from the collimator is incident on one face of the prism and emerges out from the other face. The telescope is rotated to catch the refracted image of the yellow slit. The prism table is rotated in such a direction so that the refracted images move towards the direct beam. The telescope is rotated carefully to the image in the field of view. At one stage, the image stops momentarily and turns back. This is the position of the minimum deviation as shown in the figure.

The telescope is rotated and made to coincide with the violet slit. The telescope is fixed in this position and refracted ray reading of the telescope is noted. The experiment is repeated for red slit. The prism is removed and the direct reading for the slit is taken. The difference between the direct reading and the refracted ray reading corresponding to the minimum deviation gives the angle of minimum deviation ' D '.

Using angle of the prism and angle of minimum deviation the refractive index and the dispersive power of the given prism are calculated using the above formula.

To find the angle of the prism: $TR = MSR + (VSC \times LC)$ $LC = 1'$

Position of Telescope	Telescope Reading			Telescope Reading			Difference (deg)		$A = \frac{R_1 + R_2}{4}$
	Vernier A			Vernier B			V_A	V_B	
	MSR (deg)	VSC (div)	TR (deg)	MSR (deg)	VSC (div)	TR (deg)	R_1	R_2	
LEFT									
RIGHT									

Determination of Least Count of a Spectrometer:

$$\text{Least Count} = (1MSD - 1VSD)$$

$$2 \text{ MSD} = 1^\circ$$

$$\therefore 1 \text{ MSD} = \left(\frac{1}{2}\right)^\circ$$

$$30 \text{ VSD} = 29 \text{ MSD}$$

$$1 \text{ VSD} = \left(\frac{29}{30}\right) \text{ MSD}$$

$$1 \text{ VSD} = \left(\frac{29}{30}\right) \times \left(\frac{1}{2}\right)^\circ$$

$$1 \text{ VSD} = \left(\frac{29}{60}\right)^\circ$$

$$\text{Least count} = \left(\frac{1}{2}\right)^\circ - \left(\frac{29}{60}\right)^\circ$$

$$\text{Least count} = \left(\frac{30 - 29}{60}\right) = \left(\frac{1}{60}\right)^\circ = 1'$$

$$\therefore \text{Least count} = 1'$$

To find the Refractive Index

$$TR = MSR + (VSC \times LC)$$

LC= 1'

[illegible]

To find the Dispersive Power of a Prism (ω):

Sl.No	Colour I	μ_1	Colour II	μ_2	$\mu = \left(\frac{\mu_1 + \mu_2}{2} \right)$	$\omega = \frac{\mu_2 - \mu_1}{\mu - 1}$
1	Violet		Red			

RESULT:

- (i) The angle of the given prism is found to be $A = \text{-----}$ (deg)
- (ii) The dispersive power of the given prism for the marked lines of mercury spectrum is determined and are tabulated.

Viva Questions:

1. What are the preliminary adjustments of a Spectrometer?

- i. Fixing telescope by focusing a distant object,
- ii. Adjusting collimator along with telescope for a clear and thin slit.
- iii. Adjusting main scale and vernier scales to read zero coincidence.
- iv. Leveling the spectrometer using spirit level.

2. What will happen if white light passes through a prism?

It gets dispersed and give spectrum of colours in the other side.

3. Define Dispersion of light.

When a white light is passed through a prism, the other side the constituent colours of white light are splitted with different angle of deviations from the direct ray.

4. Whether Mercury vapour lamp is monochromatic or polychromatic light source.

Mercury vapour lamp is a polychromatic light source

5. Distinguish between Monochromatic and Polychromatic light.

Source containing single wavelength is monochromatic and source consists of many wavelength is Polychromatic.

6. What is the least count of a Spectrometer?

$$LC = 1'$$

7. What is the least value of a main scale of a spectrometer other than Zero?

$$30'$$

8. What is meant by angle of Minimum Deviation?

The angle at which particular wavelength of light finds minimum deviation from the direct ray.

9. Define Dispersive Power of a prism.

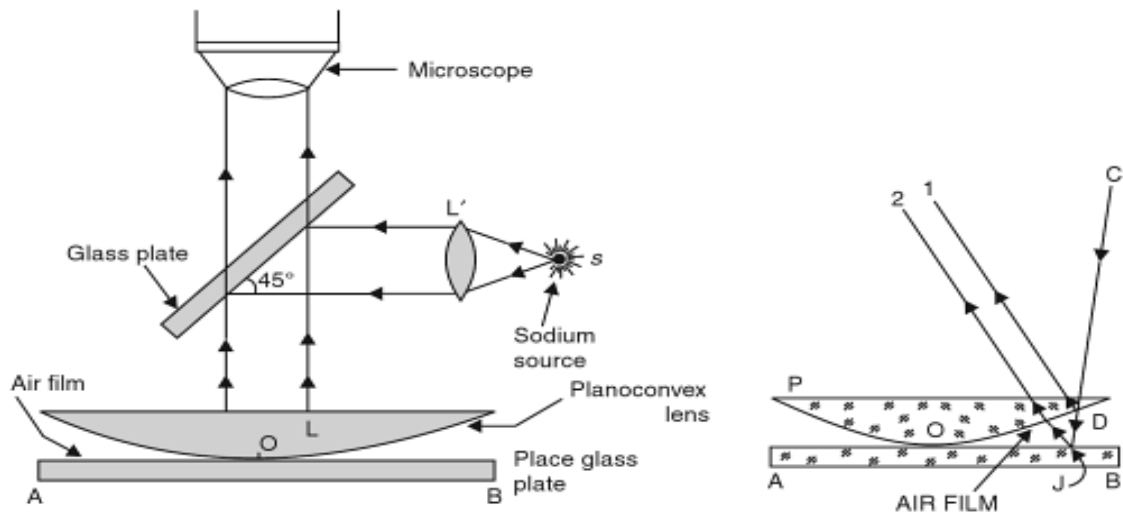
Ability of a prism to show the separation between closely placed wavelengths of a source.

10. Differentiate between Dispersive power and resolving power of a prism.

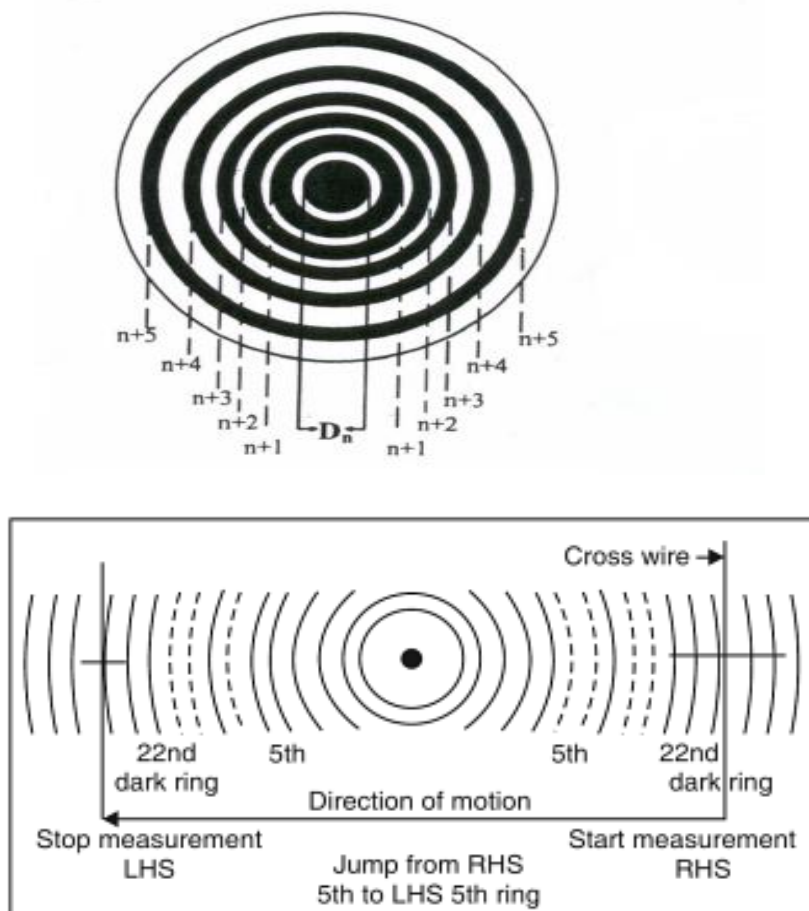
Dispersive power: Ability of a prism to show the separations between closely placed wavelengths of a source.

Resolving Power: ability of a device to show the spacing between two closely placed wavelengths of a light.

Diagram: Newton's Ring set up



Fringe Pattern:



NEWTON'S RINGS

Expt No:

Date:

AIM:

To determine the radii of curvature of the two faces of a given bi-convex lens by forming Newton's ring and to evaluate the focal length of the given lens.

APPARATUS REQUIRED:

A bi-convex lens, Sodium Vapour lamp, Glass Plate, Reading Lens, Travelling Microscope etc.,

FORMULA:

Radius of curvature of the first surface of the lens

$$R = \frac{D_{(m+n)}^2 - D_n^2}{4m\lambda} \dots\dots\dots (\text{m})$$

Focal length of the lens

$$f = \frac{1}{(\mu - 1) \left[\frac{2}{R} \right]} \dots\dots\dots (\text{m})$$

m – Difference between the orders of the fringe

λ - Wavelength of the monochromatic source (**5893 Å**)

μ - Refractive index of the lens material. (1.5)

$D_{(m+n)}$ - Diameter of the (m+n)th ring (**m**)

D_n - Diameter of the nth ring (**m**)

R-Radius of curvature of the first surface (**m**)

PROCEDURE:

The apparatus is arranged as shown in the figure. The bi-convex of large focal length is placed on the glass plate. Light from the sodium vapour lamp is directed horizontally on to the glass plate, which is kept inclined at an angle of 45° to the horizontal, by means of a condensing lens of focal length 10 to 15 cm. The light reflected by the glass plate travels vertically downwards and is incident normally on the lens and the glass plate.

Interference takes place between the portions of light reflected at the top and bottom surfaces of the air film enclosed between the lens and the glass plate. At the point of contact of the lens and the glass plate, the thickness of the air film is zero, and a

dark spot is obtained. As we move away from this point, the thickness of the film increases and alternate bright and dark fringes are obtained.

The surfaces of the glass plate and the lens are thoroughly cleaned. The apparatus is set up and the microscope is focused on the rings formed within the air film. Starting with the central dark spot, the microscope is moved to one side working its screw, so that one of its cross-wires is tangential to the $n+30^{\text{th}}$ dark ring. The readings of the scale and vernier of the microscope are taken. The microscope is then adjusted so that the cross-wire is tangential to the $n+27^{\text{th}}$ dark ring and the reading of microscope is taken. Similarly the readings corresponding to the $n+24^{\text{th}}$, $n+21^{\text{th}}$, $n+18^{\text{th}}$ etc., rings are taken. The microscope is then moved to the other side of the central spot and the corresponding readings are taken. The diameters of the different rings are determined and using the wavelength of sodium light, the radius of curvature of the face in contact with the glass plate is determined.

The experiment is repeated keeping the other face of the lens in contact with the glass plate and the radius curvature of this face is also calculated. Assume the wavelength of the sodium vapour lamp and the refractive index of the given lens. Then the focal length of the given lens is determined.

Least Count of Travelling Microscope:

Least count = 1MSD – 1VSD

20MSD = 1cm

Value of 1MSD = $\left(\frac{1}{20}\right) \text{cm}$

Number of vernier scale division VSD = 50

$\therefore 50\text{VSD} = 49\text{MSD}$

$\therefore 1\text{VSD} = \left(\frac{49}{50}\right) \text{MSD} = \left(\frac{49}{50}\right) \times \left(\frac{1}{20}\right) \text{cm} = \left(\frac{49}{1000}\right) \text{cm}$

$$\begin{aligned} \text{LC} &= \left(\frac{1}{20}\right) \text{cm} - \left(\frac{49}{1000}\right) \text{cm} \\ &= \left(\frac{50 - 49}{1000}\right) \text{cm} = \left(\frac{1}{1000}\right) \text{cm} \end{aligned}$$

$\therefore \text{LC} = 0.001 \text{cm}$

RESULT:

- (i) The radius of curvature of the first surface of the given lens $R = \text{----- m}$
- (ii) The focal length of the given bi-convex lens $f = \text{----- m}$

Viva Questions:

1. Give the basic principle involved in Newton's rings experiment.

Interference is the basic principle involved in Newton's rings experiment

2. Define Interference.

Superposition of two lights of same wavelength with suitable path difference results in alternative bright and dark bands of equal width is called as Interference.

3. What are the conditions to be satisfied for the phenomena of Interference?

- (i) The two sources of light should emit continuous waves of same wavelength and same time period i.e. the source should have phase coherence.
- (ii) The two sources of light should be very close to each other.
- (iii) The waves emitted by two sources should either have zero phase difference or no phase difference.

4. Define Coherence.

Those sources of light which emit light waves continuously of same wavelength, and time period, frequency and amplitude and have zero phase difference or constant phase difference are coherent sources

5. Name the experiment which proved the phenomena of interference.

Young's double slit experiment.

6. Define Superposition principle.

If amplitudes of two lights are super imposed on each other the resultant amplitude is the sum of amplitudes of two light waves.

7. What is meant by Radius of Curvature?

The distance from the center of a circle or [sphere](#) to its surface is its [radius](#)

8. Define Focal length of a lens?

It is the distance over which initially [collimated](#) rays are brought to a [focus](#)

9. What are the two type of interference?

Constructive interference and destructive interference.

10. Define path difference

It is the difference in path traversed by the two waves, measure in terms of wavelength of the associated wave.

11. Define phase difference.

It is the difference between some reference points in 2 waves. It's how much one wave is shifted from the other.

12. What is the relation between path and phase difference?

Phase difference (δ) = Wave number * Path difference

(Wave number = $2\pi / \lambda$)

13. What are the two types of interference when light falls on a thin film?

Constructive interference and destructive interference.

14. By looking the Interference pattern of Newton's rings, how will you verify whether the pattern is got due to transmitted or reflected light system?

With reflected light the central spot will be dark while on the other hand with transmitted light it will be bright

16. Why do the rings get closer as the order of the ring increases?

Moving outwards from one dark ring to the next, for example, increases the path difference by the same amount, λ , corresponding to the same increase of thickness of the air layer, $\lambda/2$. Since the slope of the convex lens surface increases outwards, separation of the rings gets smaller for the outer rings.

17. What would be effect of using white light instead of monochromatic light in Newton's rings experiment?

Coloured fringes can be seen.

18. Why is it necessary to use a lens having a large value of radius of curvature?

Large number of fringes can be seen in the field of view.

VIBRATIONAL MAGNETOMETER

Expt No:

Date:

AIM:

To determine the moment of inertia and the ratio of magnetic moments of the given two bar magnets using the Vibrational Magnetometer.

APPARATUS REQUIRED:

Vibrational Magnetometer, Two bar magnets, Stop Clock, Vernier Caliper, Screw Gauge, etc.,

FORMULA:

(i) The Moment of inertia of the first magnet

$$I_1 = m_1 \left[\frac{(L_1^2 + B_1^2)}{12} \right] \dots\dots\dots (\text{Kg m}^2)$$

(ii) The Moment of inertia of the second magnet

$$I_2 = m_2 \left[\frac{(L_2^2 + B_2^2)}{12} \right] \dots\dots\dots (\text{Kg m}^2)$$

Where,

m₁- Mass of the first bar magnet (**Kg**)

m₂- Mass of the second bar magnet (**Kg**)

L₁- Length of the first magnet (**m**)

L₂- Length of the second magnet (**m**)

B₁- Breadth of the first magnet (**m**)

B₂ - Breadth of the second magnet (**m**)

(ii) The ratio of Magnetic Moments

$$\left(\frac{m_1}{m_2} \right) = \left(\frac{I_1}{I_2} \right) \left(\frac{T_2^2}{T_1^2} \right) \text{ (No Unit)}$$

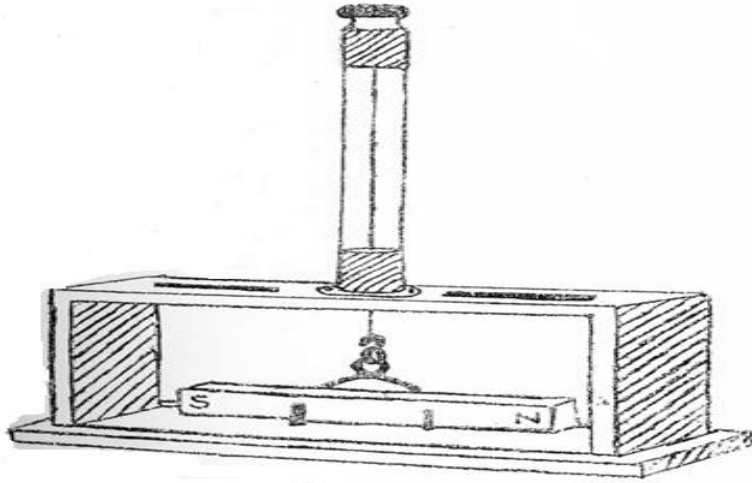
Where,

I₁ and I₂ are the Moment of inertia of the first and second magnet

T₁ and T₂ are Time period for one oscillation for the first and second magnet.

Diagram:

Vibrational Magnetometer



1. To find the time period:

Sl.No	Magnets	Mass of the Magnets $\times 10^{-3}\text{Kg}$	No: of oscillations	Time period sec	Time period for one oscillation sec	Average time period sec
1	I		0-20			(T ₁)
2			0-25			
3			0-30			
4	II		0-20			(T ₂)
5			0-25			
6			0-30			

PROCEDURE:

A strip is freely suspended within a rectangular wooden box by means of a spun silk fibre. The rectangular box is provided with glass slides, which can be opened. A plane mirror is fixed on the inside bottom, and a line is drawn centrally on it parallel to the length. The given magnet is placed on the strip. The box is adjusted for leveling using a spirit level then the rectangular box is adjusted such that the axis of the magnet is in line with the observer's eye and the vertical line on the plane mirror.

The suspended magnet is made to oscillate by bringing a magnet near the box on one side. The oscillations are in the horizontal plane without any wobbling. By the aid of a stop clock the time taken for 20, 25 and 30 oscillations is successively noted, and the average period is calculated. The mass of the magnets are determined using a balance. The length and breadth of the magnet is measured using a vernier caliper and a screw gauge. The moment of inertia and the ratio of magnetic moments of two bar magnets are calculated using the above formula.

Least Count of a Vernier Caliper

$$\text{Least count} = 1\text{MSD} - 1\text{VSD}$$

$$10\text{MSD} = 1\text{cm}$$

$$\text{Value of } 1\text{MSD} = \left(\frac{1}{10}\right)\text{cm}$$

$$\text{Number of vernier scale division VSD} = 10$$

$$\therefore 10\text{VSD} = 9\text{MSD}$$

$$\therefore 1\text{VSD} = \left(\frac{9}{10}\right)\text{MSD} = \left(\frac{9}{10}\right) \times \left(\frac{1}{10}\right)\text{cm} = \left(\frac{9}{100}\right)\text{cm}$$

$$\begin{aligned}\text{LC} &= \left(\frac{1}{10}\right)\text{cm} - \left(\frac{9}{100}\right)\text{cm} \\ &= \left(\frac{10-9}{100}\right)\text{cm} = \left(\frac{1}{100}\right)\text{cm}\end{aligned}$$

$$\therefore \text{LC} = 0.01\text{cm}$$

1. To find the Length of First Magnet (L_1) using Vernier Caliper :

Zero Error ZE = ----- div

Least Count = 0.01cm

Zero Correction ZC = ----- div

Sl.No.	MSR cm	VSC div	TR = MSR+(VSCXLC) cm
1			
2			
3			
4			
5			
			Mean = ----- $\times 10^{-2}$ m

1. To find the Length of Second Magnet (L_2) using Vernier Caliper :

Zero Error ZE = ----- div

Least Count = 0.01cm

Zero Correction ZC = ----- div

Sl.No.	MSR cm	VSC div	TR = MSR+(VSCXLC) cm
1			
2			
3			
4			
5			
			Mean = ----- $\times 10^{-2}$ m

2. To find the Breadth of First Magnet (B_1) using Screw Gauge :

Zero Error ZE = ----- div

Least Count = $0.01 \times 10^{-3} \text{m}$

Zero Correction ZC = ----- div

Sl.No.	PSR mm	HSC div	Observed reading O.R.= PSR+(HSCXLC) mm	Correct Reading C.R= O.R. $\pm$ (ZCXLC) mm
1				
2				
3				
4				
5				
				Mean = ----- $\times 10^{-3} \text{m}$

3. To find the Breadth of Second Magnet (B_2) using Screw Gauge :

Zero Error ZE = ----- div

Least Count = $0.01 \times 10^{-3} \text{m}$

Zero Correction ZC = ----- div

Sl.No.	PSR mm	HSC div	Observed reading O.R.= PSR+(HSCXLC) mm	Correct Reading C.R= O.R. $\pm$ (ZCXLC) mm
1				
2				
3				
4				
5				
				Mean = ----- $\times 10^{-3} \text{m}$

Least count of a Screw Gauge:

$$\text{Least Count} = \frac{\text{Pitch}}{\text{No. of divisions on head scale}}$$

$$\begin{aligned}\text{Pitch} &= \frac{\text{No. of divisions moved in the Pitch Scale}}{\text{No. of rotations given on the Head Scale}} \\ &= \frac{4\text{mm}}{4}\end{aligned}$$

$$= 1\text{mm}$$

$$\text{Least Count} = \frac{1\text{mm}}{100}$$

$$\text{LC} = 0.01\text{mm}$$

RESULT:

(i) The Moment of Inertia of the First Magnet $I_1 = \text{----- Kg.m}^2$

(ii) The Moment of Inertia of the Second Magnet $I_2 = \text{----- Kg.m}^2$

(iii) The ratio of Magnetic Moments of two bar magnets $\left(\frac{M_1}{M_2} \right) = \text{----- (no unit)}$

Viva Questions:

1. Define Moment of Inertia.

The sum of the products of mass and square of the perpendicular distance to the axis of rotation of each particle in a body rotating about an axis.

2. Define Inertia.

The property of matter by which it retains its state of rest or its velocity along a straight line so long as it is not acted upon by an external force.

3. Define Magnetic moment of a bar magnet.

The magnetic moment of a magnet is a quantity that determines the torque it will experience in an external magnetic field.

4. What is the relation between frequency and time period?

Inversely related.

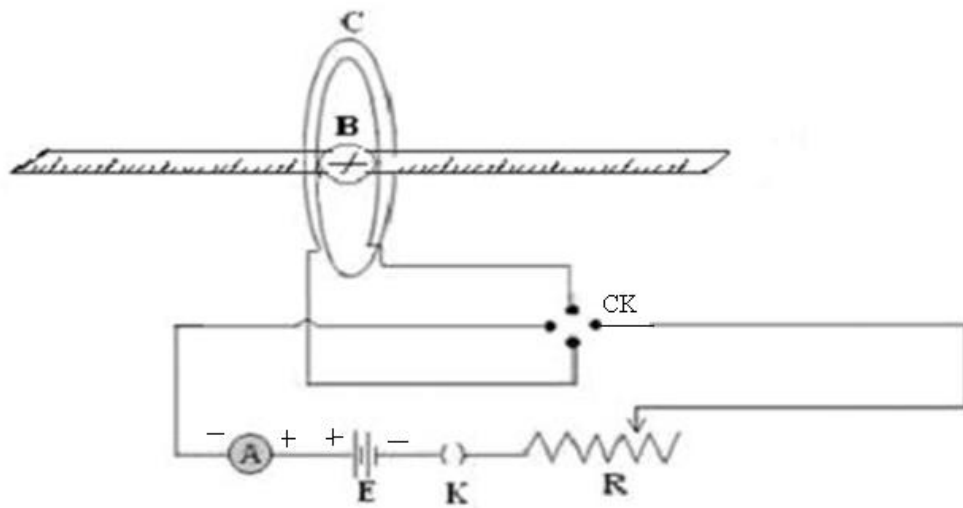
5. Define Couple or Torque.

The moment (or torque) of a force about a turning point is the force multiplied by the perpendicular distance to the force from the turning point.

6. What is meant by Magnetic meridian?

The magnetic meridian is an equivalent imaginary line connecting the magnetic [south](#) and [north poles](#) and can be taken as the magnetic force lines along the surface of the earth

CIRCUIT DIAGRAM:



C – circular coil

B – Compass box

R - rheostat

A – ammeter

E - Cell

CK- Commutator

K- Key

MAGNETIC FIELD ALONG THE AXIS OF A CIRCULAR COIL CARRYING CURRENT BY DEFLECTION MAGNETOMETER

Expt No:

Date:

AIM:

To determine the horizontal component of earth's magnetic intensity H, along the axis of a circular coil carrying current, using the deflection magnetometer.

APPARATUS REQUIRED:

Field along the axis of the coil apparatus, deflection magnetometer, accumulator, ammeter, commutator, rheostat, plug key, connecting wires, etc.,

FORMULA:

The horizontal component of earth's magnetic intensity H, along the axis of a circular coil carrying current

$$H = \left[\frac{nr^2i}{2(r^2 + x^2)^{3/2}} \right] \times \left(\frac{1}{\tan \theta} \right) \text{ampturnsm}^{-1}$$

The horizontal component of earth's magnetic field B, along the axis of a circular coil carrying current

$$B = \mu_0 H \quad \text{Tesla}$$
$$(\mu_0 = 4\pi \times 10^{-7} \text{ henry})$$

Where,

H – Horizontal component of earth's magnetic intensity (AT/m)

B– Horizontal component of earth's magnetic field (Tesla)

n - No. of turns in the circular coil

r – Radius of circular coil (m)

i – Current passed through the coil (amp)

x – Distance between centre of coil and centre of the magnetic needle (m)

θ – Deflection produced in the deflection magnetometer (deg)

PROCEDURE:

The circuit connections are made as shown in the figure. The rheostat is kept far away from the magnetometer, the deflection magnetometer is placed on the arm and adjusted for Tan A position ie, the aluminum pointer faces east west direction. The centre of the coil is made to coincide with the centre of the magnetometer without disturbing the preliminary adjustments. The arms are slightly adjusted so that the aluminum pointer reads zero-zero. Now the plane of the coil is in the magnetic meridian.

The circuit is closed by pressing the key. The number of turns in the circular coil is noted. The rheostat is adjusted for a known current, so that the deflection in the magnetometer lies between 30° to 60°. The readings at both the ends of the pointer of the magnetometer are noted, say θ_1 and θ_2 . The current is reversed by using the commutator and the readings θ_3 and θ_4 are noted. The average value θ is determined. Then the compass box is placed at some distance from the centre of the coil. The current in the circuit is adjusted using the rheostat. The readings at both the ends of the

To determine horizontal component of earth's magnetic intensity (H)

Circumference of the coil $2\pi r =$ $\times 10^{-2}\text{m}$

Radius of the coil (r) = $\times 10^{-2}\text{m}$

Sl.No	Current in Amperes	Distance between centre of coil and to the needle (x) cm	Deflection in Magnetometer (degrees)								Mean θ	tan θ	$H = \left[\frac{nr^2i}{2(r^2 + x^2)^{3/2}} \right] \times \left(\frac{1}{\tan \theta} \right)$ AT/m
			Left				Right						
			Current in one direction		Current in reverse direction		Current in one direction		Current in reverse direction				
			θ_1	θ_2	θ_3	θ_4	θ_5	θ_6	θ_7	θ_8			
1	1	2 4 6											
2	1.5	2 4 6											
3	2.0	2 4 6											
												Mean = ----- AT/m	

magnetometer needle are noted. The current direction is reversed and deflection θ_3 and θ_4 are noted. The magnetometer is placed at the same distance on the other side of the coil, the deflection θ_5 and θ_6 are noted. When the current direction is reversed θ_7 and θ_8 are tabulated. The mean value θ of the eight deflections is found. The experiment is repeated by varying distances and currents.

The circumference of the coil is found using a thread and radius is calculated. The horizontal component of earth's magnetic field is determined using the above mentioned formula.

RESULT:

The horizontal component of earth's magnetic intensity was found to be $(H) = \text{-----AT/m}$
The horizontal component of earth's magnetic field was found to be $(B) = \text{-----Tesla}$

Viva Questions:**1. What will happen when current flows through a coil?**

The deflection in the magnetometer lies between 30° to 60° . The readings at both the ends of the pointer of the magnetometer are noted, say θ_1 and θ_2 . The current is reversed by using the commutator and the readings θ_3 and θ_4

2. What are Tan A and Tan B positions?

Tan-A the aluminum pointer faces east west direction Tan-B the aluminum pointer faces north and south direction

3. Define magnetic meridian?

The Aluminium pointer faces east west direction. The centre of the coil is made to coincide with the centre of the magnetometer without disturbing the preliminary adjustments. The arms are slightly adjusted so that the aluminum pointer reads zero-zero. Now the plane of the coil is in the magnetic meridian.

4. Define Tangent law.

$F = H \tan \theta$ H-Horizontal component of earth's magnetic intensity (AT/m)

θ – Deflection produced in the deflection magnetometer (deg)

5. Define magnetic moment.

$M = i a$ Where I is the electric current that flows through a circular wire of an area of cross section 'a'

THERMAL CONDUCTIVITY OF A BAD CONDUCTOR (LEE'S DISC METHOD)

Expt No:

Date:

AIM:

To determine the thermal conductivity of a bad conductor using Lee's disc apparatus

APPARATUS REQUIRED:

Lee's disc apparatus, bad conductor, Thermometer, Stop Clock, Steam Boiler, Screw Gauge, Vernier Caliper, etc.,

FORMULA:

Thermal Conductivity of a bad conductor is given by

$$K = \left[\frac{MS \left(\frac{d\theta}{dt} \right)_{\theta_2} d}{\pi r^2 (\theta_1 - \theta_2)} \right] \times \left[\frac{r + 2h}{2r + 2h} \right] \text{ w/m/k}$$

Where,

M – Mass of the metallic disc in Kg

S – Specific heat capacity of the material of the disc (J/Kg/K)

$\left(\frac{d\theta}{dt} \right)_{\theta_2}$ - Rate of cooling at steady temperature θ_2 (°C / sec)

θ_1 – Steady temperature of the steam chamber (°C)

θ_2 – Steady temperature of the metallic disc (°C)

r – Radius of the metallic disc (m)

h- Thickness of the metallic disc (m)

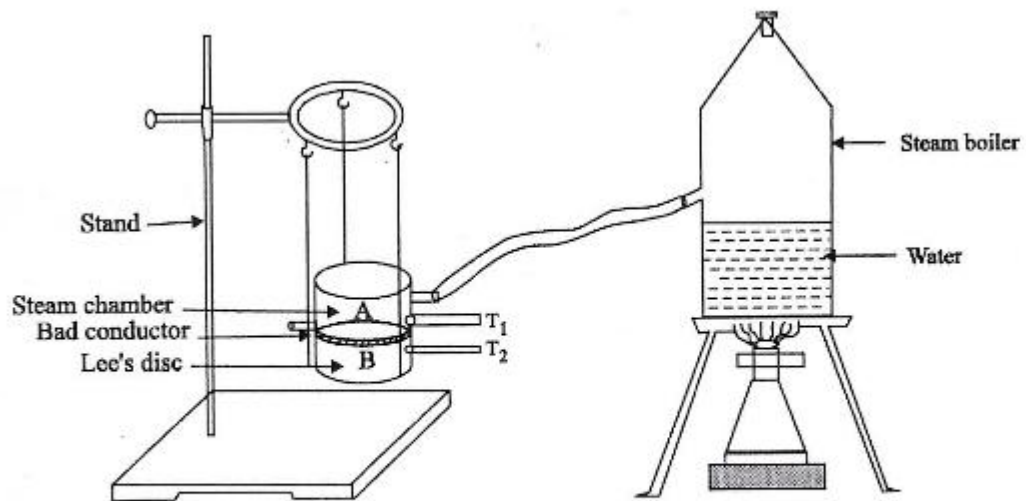
d – Thickness of the bad conductor (m)

PROCEDURE:

The thickness of the bad conductor and the thickness of the metallic disc are found using screw gauge. The radius of the metallic disc is found using vernier caliper. The mass of the metallic disc is also found using a common balance. The reading is tabulated.

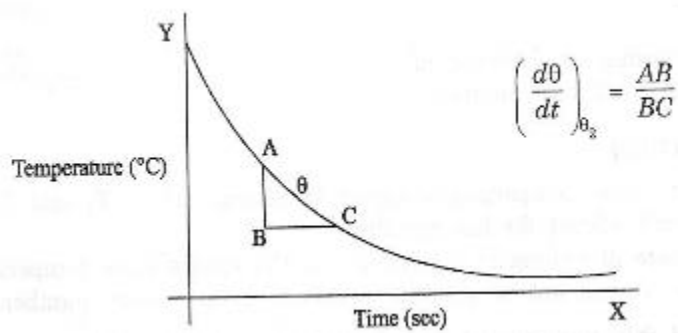
The whole apparatus is suspended from a stand as shown in the figure. The given bad conductor is placed in between the metallic disc and steam chamber. Two thermometers T_1 and T_2 are inserted in the respective holes, steam from the steam boiler is passed into the steam chamber until the temperature of the steam chamber and the metallic disc are steady. The steady temperature of the steam chamber θ_1 and steady temperature of the metallic disc θ_2 recorded by the thermometer are noted.

Diagram:



Lee's disc apparatus

Model Graph:



Now the cardboard is removed and the steam chamber is placed in direct contact with the metallic disc. The temperature of the disc rapidly rises. When the temperature of the disc raises about 10°C above θ_2 (Steady temperature of the metallic disc), the steam chamber is carefully removed, after cutting off the steam supply. When the temperature of the disc reaches 5°C above the steady temperature of the disc that is $(\theta_2 + 5^{\circ}\text{C})$ a stop clock is started. Time for every 1°C fall of temperature is noted until the metallic disc attains temperature $(\theta_2 - 5^{\circ}\text{C})$.

Graph:

A graph is drawn taking time along the X- axis and temperature along Y-axis. To obtain the rate of cooling $(d\theta/dt)$ from this graph, a triangle is drawn by taking 0.5°C above and 0.5°C below the steady temperature θ_2 . Then the slope AB / BC gives the rate of cooling $(d\theta/dt)$ at θ_2 . From these reading and using the given formula the thermal conductivity of the given cardboard is calculated.

Least Count of a Vernier Caliper

Least count = $1\text{MSD} - 1\text{VSD}$

$10\text{MSD} = 1\text{cm}$

Value of $1\text{MSD} = \left(\frac{1}{10}\right)\text{cm}$

Number of vernier scale division $\text{VSD} = 10$

$\therefore 10\text{VSD} = 9\text{MSD}$

$\therefore 1\text{VSD} = \left(\frac{9}{10}\right)\text{MSD} = \left(\frac{9}{10}\right) \times \left(\frac{1}{10}\right)\text{cm} = \left(\frac{9}{100}\right)\text{cm}$

$\text{LC} = \left(\frac{1}{10}\right)\text{cm} - \left(\frac{9}{100}\right)\text{cm}$

$= \left(\frac{10-9}{100}\right)\text{cm} = \left(\frac{1}{100}\right)\text{cm}$

$\therefore \text{LC} = 0.01\text{cm}$

Determination of Rate of Cooling of Metallic Disc $\left(\frac{d\theta}{dt}\right)$

Steady temperature of the steam chamber $\theta_1 = \text{-----}$ °C

Steady temperature of the metallic disc $\theta_2 = \text{-----}^\circ\text{C}$

Sl.No.	Temperature °C	Time sec

1. To find the Thickness of the Cardboard (d) using Screw Gauge :

Zero Error ZE = ----- div

Least Count = $0.01 \times 10^{-3} \text{m}$

Zero Correction ZC = ----- div

Sl.No.	PSR mm	HSC div	Observed reading O.R.= PSR+(HSCXLC) mm	Correct Reading C.R= O.R. ± (ZCXLC) mm
1				
2				
3				
4				
5				
				Mean = ----- X10 ⁻³ m

2.To find the Thickness of the Metallic Disc (h) using Screw Gauge :

Least Count = $0.01 \times 10^{-3} \text{m}$

Zero Error ZE = ----- div

Zero Correction ZC = ----- div

Sl.No.	PSR mm	HSC div	Observed reading O.R.= PSR+(HSCXLC) mm	Correct Reading C.R= O.R. $\pm$ (ZCXLC) mm
1				
2				
3				
4				
5				
				Mean = ----- $\times 10^{-3} \text{m}$

3. To find the Diameter of the Metallic Disc (D) using Vernier Caliper :

Least Count = 0.01cm

Sl.No.	MSR cm	VSC div	TR = MSR+(VSCXLC) cm
1			
2			
3			
4			
5			
			Mean = ----- $\times 10^{-2} \text{m}$

RESULT:

The thermal conductivity of the given bad conductor is determined by the Lee's disc method $K = \text{----- W/m/K}$

Viva Questions:

1. Differentiate Conduction, Convection and Radiation.

Conduction: It is the process in which heat is transferred from region of higher temperature to the region of lower temperature of body without any transfer of matter or molecules

Convection: It is the process in which particles of the material at higher temperature move to cold region and particles at low temperature move to hotter region setting up a cyclic flow of material.

Radiation: It is the process in which heat is transferred by electromagnetic waves of wavelength from about 7500 Å to about 3mm. these waves do not need any material medium.

2. Classify materials based on the property of conduction.

In non-metallic solids, liquids, gases, such collisions being few, heat transfer is slow process, hence these are bad conductor.

In metals, free electrons are present which move rapidly and collide very frequently, thus transferring heat. Due to free electrons, metals are good conductors of electricity.

3. Differentiate between Conductor, Semiconductor and Insulator.

Conductor

1. The conductor is the material which has low resistivity

2. The conduction band and valence band overlaps and there is no forbidden band gap.

Semiconductor

1. The semiconductor is the material which has resistivity lying between a conductor and an insulator

2. The conduction band and valence band is separated by a forbidden band gap of energy Eg.

Insulator.

1. Insulator is the material which has zero resistivity

2. The conduction band and valence band is separated by a large distance forbidden band gap of energy Eg.

4. Why conductors are good in conducting heat and electricity.

1. The electrical and thermal conductivity decreases with the increase in temperature and impurities

2. The electrical and thermal conductivity is very high at low temperature

3. For non-metal, the electrical and thermal conductivity is very less

5. Define specific heat capacity.

The quantity of heat required to raise the temperature of unit mass of the substance through 1 kelvin.

$$S = \frac{Q}{m\theta} \text{ J/kg/K}$$

6. Define Co-efficient of thermal conductivity.

The quantity of heat conducted per unit area per unit time maintained at unit temperature gradient

7. Define temperature gradient.

The rate of fall of temperature with respect to distance is called temperature gradient.

The smaller values $\frac{\theta_1 - \theta_2}{x}$ is written as $\frac{d\theta}{dx}$. It is known as the temperature gradient.

8. What is meant by steady temperature?

The rate of flow of heat is same throughout the material.

SPECTROMETER – (Wavelength of Mercury source using Grating)

Expt No:

Date :

AIM:

To determine the wavelengths of Mercury vapour lamp using diffraction grating by Normal Incidence method.

APPARATUS REQUIRED:

Spectrometer, Diffraction Grating, Mercury Vapour Lamp, Reading Lens etc.,

FORMULA:

No. of lines in the grating,
$$N = \frac{\sin \theta}{m\lambda} \dots\dots\dots (\text{Lines/m})$$

Wavelength of the line,
$$\lambda = \frac{\sin \theta}{mN} \dots\dots\dots (\text{\AA})$$

Where,

N- Number of lines in the grating (lines/m)

θ – Angle of diffraction for normal incidence (deg)

m – Order of the spectrum

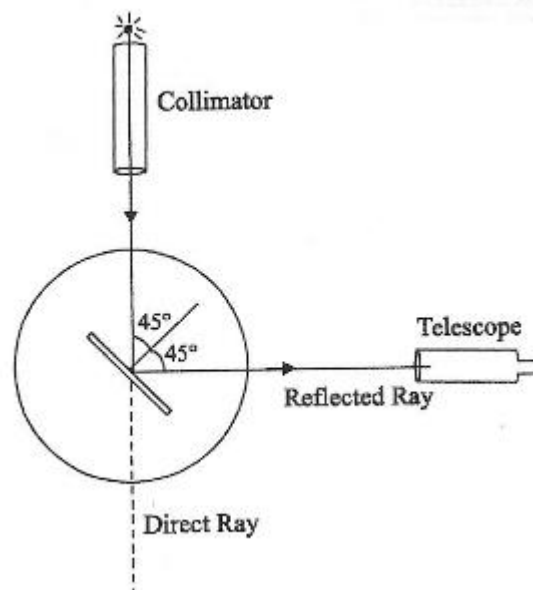
λ – Wavelength of the source ($\text{\AA}$)

PROCEDURE:

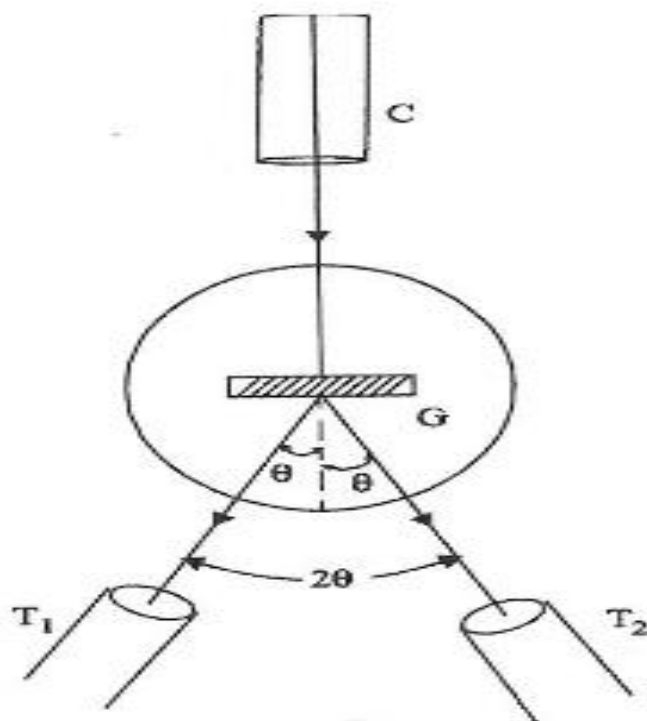
1. Adjustments of the grating for Normal incidence

The initial adjustments of the spectrometer are made as usual. The vernier table is adjusted as $0^\circ - 180^\circ$ after making the vertical cross wire to coincide with the fixed edge of the image of the slit, which is illuminated by source of light. Then the telescope is rotated by an angle 90° (right hand side) and fixed. The grating table is rotated until on seeing through the telescope the reflected image of the slit coincides with the vertical cross-wire. This is possible only when a light emerging out from the collimator is incident at an angle of 45° towards the collimator. Now light coming out from the collimator will be incident normally on the grating.

Diagram:
Normal Incident Position



Spectrometer Grating



2. To find the Number of lines per metre of Grating (N)

The slit is illuminated by a source of known wavelength. The telescope is released to catch the diffracted image of the first order on the left side and coincide with the vertical crosswire. The readings in two verniers are noted, then the telescope is turned to right side to catch the diffracted image of the first order and coincide with the vertical crosswire, the readings are noted. The difference between the positions of the right and left sides readings gives the twice the angle of diffraction 2θ . The number of lines per metre of the grating is calculated by using the given formula assuming the wavelength of the given source.

2. Determination of Wavelength of Mercury Spectrum

The telescope of the spectrometer is moved to either side of the direct image, the diffraction pattern of the spectrum of the first order and second order are seen. The reading for first order are taken by coinciding the prominent lines namely Violet I, II, Bluish Green I and II, Yellow I and II and Red I and II with the vertical crosswire coinciding with the colors. The readings are tabulated and from this the angles of diffraction for different colours are determined. The wavelengths for different colours are calculated by using the given formula.

To find the Number of Lines in the Grating: $TR = MSR + (VSC \times LC)$ $LC = 1'$

Diffracted Ray	Telescope Reading			Telescope Reading			Difference (deg)		Mean $\theta = \left(\frac{V_A + V_B}{4} \right)$	N = $\frac{\sin \theta}{m \lambda}$
	Vernier A			Vernier B			V _A	V _B		
	MSR (deg)	VSC (div)	TR (deg)	MSR (deg)	VSC (div)	TR (deg)	2θ	2θ		
LEFT										
RIGHT										

Determination of Least Count of a Spectrometer:

$$\text{Least Count} = (1\text{MSD} - 1\text{VSD})$$

$$2 \text{ MSD} = 1^\circ$$

$$\therefore 1 \text{ MSD} = \left(\frac{1}{2}\right)^\circ$$

$$30 \text{ VSD} = 29 \text{ MSD}$$

$$1 \text{ VSD} = \left(\frac{29}{30}\right) \text{ MSD}$$

$$1 \text{ VSD} = \left(\frac{29}{30}\right) \times \left(\frac{1}{2}\right)^\circ$$

$$1\text{VSD} = \left(\frac{29}{60}\right)^\circ$$

$$\text{Least count} = \left(\frac{1}{2}\right)^\circ - \left(\frac{29}{60}\right)^\circ$$

$$\text{Least count} = \left(\frac{30 - 29}{60}\right)^\circ = \left(\frac{1}{60}\right)^\circ = 1'$$

$$\therefore \text{Least count} = 1'$$

RESULT:

1. The number of lines per metre length of the grating $N = \text{-----}$ lines / m
2. The wavelengths of prominent spectral lines of mercury spectrum are found and are tabulated.

Viva Questions:

1. Define diffraction.

The phenomenon of spreading of light when it passes through a narrow aperture or bending of light around the edges of an opaque obstacle is known as diffraction of light.

2. What are the types of diffraction?

The two types of diffraction are

- ☐ Fresnel diffraction and
- ☐ Fraunhofer diffraction

3. Distinguish between Fresnel's and Fraunhofer's diffraction.

Fresnel Diffraction.

In this diffraction the source of light and the screen are separated by a finite distance. In this case the wave front is either spherical or cylindrical and the diffraction pattern is obtained without using any lens.

Fraunhofer Diffraction.

In this diffraction the source of light and screen are placed at infinite distance. To bring the diffracted ray into focus, a convex lens is placed in between the slit and the screen.

4. What is a Grating? How are they made?

Diffraction grating is constructed by drawing equidistant parallel lines on a transparent material like glass plate by means of a diamond edge. The rulings made on the glass plate act as an opaque medium for the light, whereas the spacing between any two lines act as a transparent medium for the incident light. In a plane transmission gratings nearly 1500 lines are drawn in one inch space.

5. What is meant by grating element?

The distance $a+b$ is known as grating element

6. What are different types of grating?

- i) A plane transmission gratings nearly 15000 lines are drawn in one inch space.
- ii) A when light is reflected from the surface of the grating. Such a grating is known as reflection grating

7. What is a plane transmission grating?

Diffraction grating or plane transmission grating is an arrangement consisting of a large number of parallel and equidistant slits of the same width (10^{-6}) on a transparent material such as glass, with a diamond point. The ruled lines are opaque to light while the space between any two lines is transparent to light and act as light. This is known as plane transmission grating.

8. What is a reflection grating?

When light is reflected from the surface of the grating. Such a grating is known as reflection grating

9. What will be the maximum number of orders available with a grating?

Maximum order of Spectrum is obtain.

$$\sin \theta = Nn\lambda \text{-----(2)}$$

The maximum order of spectrum in a diffraction grating is obtained by substituting $\sin \theta = 1$ in equation (2). Therefore equation (2) can be written as,

$$1 = nN\lambda$$

$$n = \frac{1}{N\lambda}$$

For a wavelength of $\lambda = 6000 \text{ \AA}$ and $N = 6 \times 10^5 \text{ lines/m}$, we get,

$$n = \frac{1}{6000 \times 10^2 \times 6000 \times 10^{-10}} \text{ (or) } n = 2.77$$

Therefore, the maximum order that can be observed is 2

10. How will you adjust the grating for Normal incidence position

The initial adjustments of the spectrometer are made as usual. The vernier table is adjusted as $0^\circ - 180^\circ$ after making the vertical cross wire to coincide with the fixed edge of the image of the slit, which is illuminated by source of light. Then the telescope is rotated by an angle 90° (right hand side) and fixed. The grating table is rotated until on seeing through the telescope the reflected image of the slit coincides with the vertical cross-wire. This is possible only when a light emerging out from the collimator is incident at an angle of 45° towards the collimator. Now light coming out from the collimator will be incident normally on the grating.

11. What are the preliminary adjustments of Spectrometer?

Preliminary adjustment of the spectrometer is made as given below.

(i) Eyepiece adjustments:

The spectrometer is turned towards a bright object say the white wall 3 and 4m away and the eyepiece is adjusted so that the crosswire are clearly seen

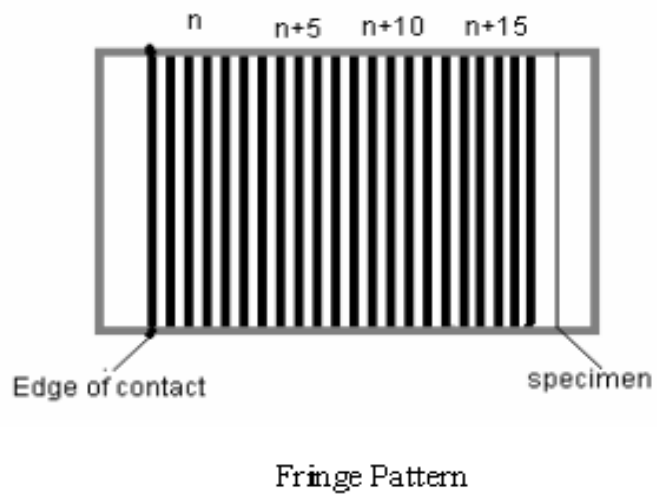
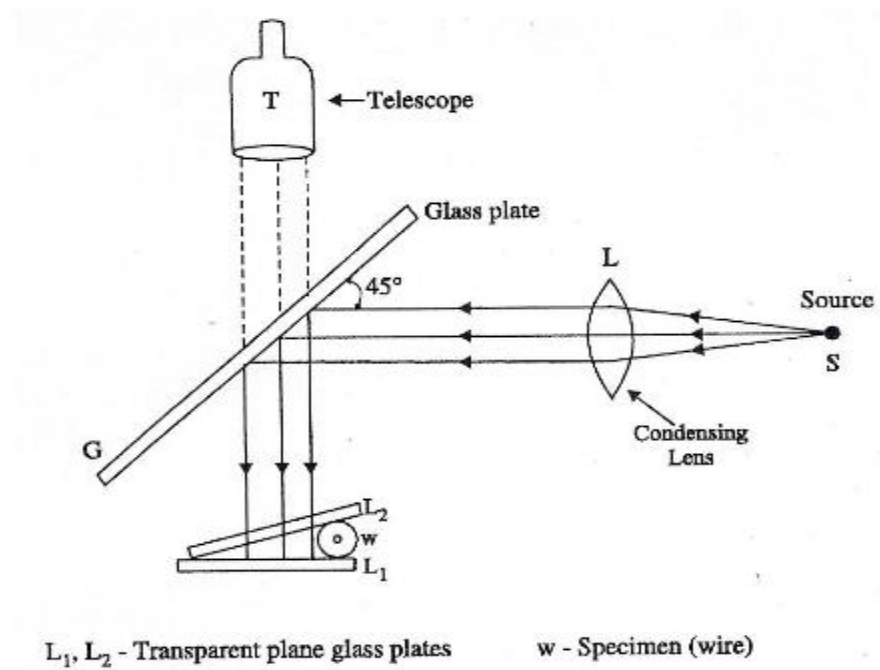
(ii) Telescope adjustments:

The telescope is turned towards a distant object and is adjusted until the distant object is clearly seen. Thus the telescope is set for receiving the parallel rays of the collimator.

(iii) Collimator Adjustments:

The slit of the collimator is illuminated with a monochromatic light. The telescope is turned to view the image of the slit and the collimator screen is adjusted such that a clear image of the slit is obtained. The slit of the collimator must be adjusted to be vertical and narrow. The initial leveling of the prism is done using a spirit level.

DIAGRAM:



AIR WEDGE

Expt No:

Date:

AIM:

To find the thickness (diameter) of a thin wire by forming interference.

APPARATUS REQUIRED:

Travelling microscope, sodium vapour lamp, two optically plane rectangular glass plates, condensing lens, reading lens and thin wire.

FORMULA:

Thickness of the wire, $t = \frac{\lambda l}{2\beta}$ (m)

λ - Wavelength of the incident light-(**5893 Å**).

l - Distance between the edge of contact & wire (m).

β - Fringe width (m).

PROCEDURE:

An air wedge is formed by keeping two optically plane glass plates in contacts along one of their edges. At the end, the thin wire is introduced with its length perpendicular to the length of the plane. The glass plates are tied together in this position by means of two rubber bands. It is then placed on the horizontal bed plate of the travelling microscope.

The interference pattern can be obtained at 45° to the horizontal plane and condensing lens(Fig 1). Light from the sodium vapour lamp is made to fall vertically on the airwedge. These interference fringes are viewed through the travelling microscope. A system of equi-spaced straight alternatively dark and bright bands is obtained (Fig 2).

The crosswire of the telescope is adjusted to coincide with the centre of the well defined dark band near the edge of contact of the glass plates. It is taken as the n^{th} band. The reading on the horizontal scale of the microscope is noted. The microscope is then moved in the same direction by working the horizontal transverse screw and made to coincide with every successive fifth fringes. The readings are noted and this is to be continued till about 50 fringes are covered. The readings are tabulated. From these readings, the mean width of one fringe (β) is found.

The distance 'l' between the edges of contact and the wire is measured with the help of the traveling microscope. Assuming the wavelength of the sodium light, the thickness of the thin wire is calculated by using the given formula.

(i) Determination of the band width (β) for specimen :

$$TR = MSR + (VSC \times LC)$$

$$\text{Least count} = 0.001 \times 10^{-2} \text{m}$$

S.No	Order of the band	Microscope reading			Width of 5 bands cm	Mean width of one band (β_1) cm
		MSR cm	VSC div	TR cm		
1.	n					
2.	n+5					
3.	n+10					
4.	n+15					
5.	n+20					
6.	n+25					
7.	n+30					
8.	n+35					
9.	n+40					
10.	n+45					
11.	n+50					
						Mean = ----- X 10^{-2}m

To find the distance between Specimen I and the edge of contact (l):

$$LC = 0.001 \text{cm}$$

Sl.No.	Position of the telescope	MSR cm	VSC div	TR = MSR + (VSC $\times$ LC) cm	$l = l_1 \approx l_2$ cm
1.	At the edge of contact				
2.	At the Specimen				

LEAST COUNT DERIVATION FOR MICROSCOPE:

Least count = 1MSD – 1VSD

20MSD = 1cm

Value of 1MSD = $\left(\frac{1}{20}\right)cm$

Number of vernier scale division VSD = 50

$\therefore 50VSD = 49MSD$

$\therefore 1VSD = \left(\frac{49}{50}\right)MSD = \left(\frac{49}{50}\right) \times \left(\frac{1}{20}\right)cm = \left(\frac{49}{1000}\right)cm$

$LC = \left(\frac{1}{20}\right)cm - \left(\frac{49}{1000}\right)cm$

$= \left(\frac{50 - 49}{1000}\right)cm = \left(\frac{1}{1000}\right)cm$

$\therefore LC = 0.001cm$

RESULT:

The thickness of the given specimens are found by forming interference fringes using air-wedge arrangement

(i) Thickness of specimen I $t_1 = \text{----- m.}$

(ii) Thickness of specimen II $t_2 = \text{----- m.}$

Viva Questions:

1. Give the basic principle involved in Air- Wedge experiment.

Interference of light

2. Define Interference.

When two light waves superimpose, then the resultant amplitude or intensity in the region of superposition is different than the amplitude of individual waves. This modification in the distribution of intensity in the region of superposition is called interference.

3. What are the conditions to be satisfied for the phenomena of Interference?

The two sources should be coherent

The two sources must emit continuous waves of the same wavelength and time period.

The separation between the two sources ($2d$) should be small

The distance D between the two sources and screen should be large

The sources must be narrow

The sources should be monochromatic

The amplitude of the interfering waves should be equal or nearly equal.

4. Define Coherence.

When the light waves are in same phase and with all most same wavelength, frequency and travelling in same direction than the light beams are called as coherent sources and this property is called as coherent

5. Name the experiment which proved the phenomena of interference.

Young's double slit experiment.

6. Define Superposition principle.

The principle states that resultant displacement of a particle of the medium acted upon by two or more waves simultaneously is the algebraic sum of the displacements of the same particle due to individual waves, in the absence of the others.

7. What are the two types of interference?

1) Constructive interference

2) Destructive interference

8. What is the relation between Path and Phase difference?

Phase difference (δ) = Wave number * Path difference

(Wave number = $2\pi / \lambda$)

9. What is the least count of traveling Microscope?

Least count = 0.001 cm

10. In what unit the main scale of traveling microscope is calibrated?

In terms of centimeter

11. What type of error we can expect while taking reading in a traveling microscope?

Back slash error

12. What is meant by Wedge shaped thin film?

When two plane glass plate of equal thickness and diameter is tied at one end, then the two plane surfaces is inclined at an angle θ . Thus a wedge shaped air film is formed between the two glass plates and hence called as an air wedge. It is used to find the thickness of very thin materials

like hair, paper etc.,

13. Define bandwidth β .

The distance between any two consecutive bright and dark fringes is known as fringe width β . It is constant for both bright and dark fringe. It is given by

$$\beta = \frac{\lambda D}{d}$$

14. List out the applications of Air wedge.

Air Wedge is used to find

To find the thickness of very thin materials like hair, paper

To check the flatness of a given surface

POLARIMETER

AIM:

To determine the specific rotatory power of cane sugar solution using Polarimeter.

APPARATUS REQUIRED:

Polarimeter and accessories, solution of cane sugar, sodium vapour lamp, distilled water etc.,

FORMULA:

$$\text{Specific rotatory power, } S = \frac{\theta}{LC} \dots\dots\dots(\text{deg} / \text{dm})$$

θ - Angle of rotation (deg)

L - Length of the tube-solution (dm)

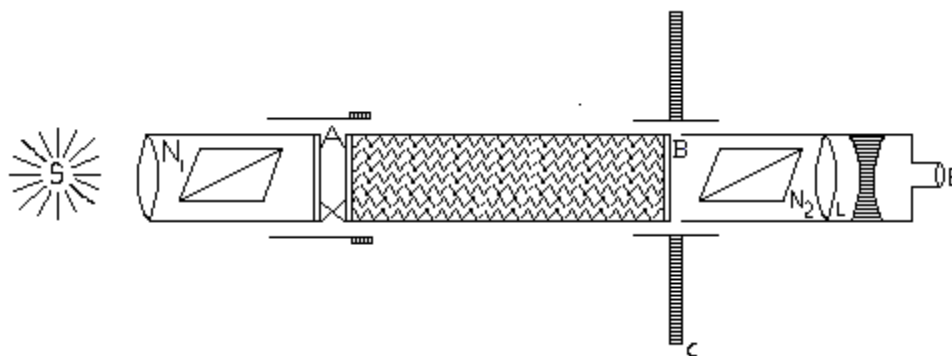
C - Mass concentration (g/cc)

PROCEDURE:

The bi-quartz polarimeter consists of two semicircular plates of quartz of the same thickness one of which produces left hand rotation, both bring out with their refracting surfaces perpendicular to the principal axis.

The circular scale and vernier are studied and the least count is determined. The eyepiece is adjusted for uniform illumination. The glass tube AB is filled with distilled water without leaving air bubbles inside and placed in position. If the analyzing nicol is set with its principal plane parallel to the plane of vibration of the incident light, both halves of the field of view will be of same colour, which is called the tint of passage. If the analyzing nicol is rotated slightly one half appears reddish and other bluish. At first with distilled water the analyzing nicol is put in 'Tint of Passage' position and the reading of main scale and vernier are taken.

A 2% solution of cane sugar is prepared by dissolving 2 grams of cane sugar in 100 ml distilled water. The solution is filtered and poured in the solution tube. This 2% solution in tube is introduced and now the colours of the two halves are found to differ due to rotation of plane of vibration by the solution. The analyzing nicol is rotated till the 'Tint of passage' is restored. The readings of the circular scale and vernier are noted. The readings of the circular scale and vernier are noted. The same adjustment is repeated and the mean reading is taken. The difference between the distilled water and 2% solution of the readings gives the angle of rotation ' θ '.



C - Circular scale

S - Source

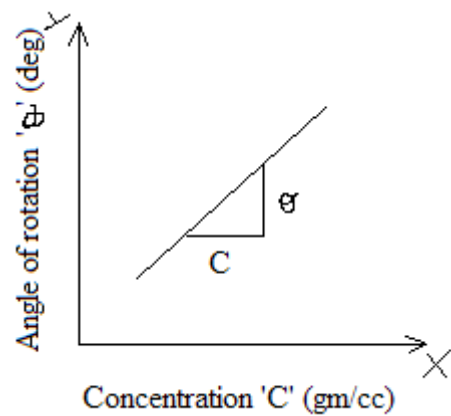
X - Biquartz

N_1 - Polarising nicol

N_2 - Analysing nicol

L - Lens

MODEL GRAPH:



And the experiment is repeated with 4%, 6%, 8% and 10% cane sugar solution and specific rotatory power 'S' is calculated using the formula

$$S = \left(\frac{\theta}{LC} \right).$$

The value of S is verified graphically by plotting the graph, concentration along X-axis and angle of rotation along Y-axis. From graph $\left(\frac{\theta}{C} \right)$ is calculated and the value substituted in the formula.

Determination of Least Count of Polarimeter:

$$\text{Least Count} = (1\text{MSD} - 1\text{VSD})$$

$$1 \text{ MSD} = 1^\circ$$

$$10 \text{ VSD} = 9 \text{ MSD}$$

$$1 \text{ VSD} = \left(\frac{9}{10} \right) \text{MSD}$$

$$1 \text{ VSD} = \left(\frac{9}{10} \right) \times (1)^\circ$$

$$1\text{VSD} = \left(\frac{9}{10} \right)^\circ$$

$$\text{Least count} = (1)^\circ - \left(\frac{9}{10} \right)^\circ$$

$$\text{Least count} = \left(\frac{10-9}{10} \right) = \left(\frac{1}{10} \right)^\circ = (0.1)^\circ$$

$$\text{Least count of polarimeter} = 0.1 \text{ deg} = 6 \text{ minute}$$

RESULT:

The specific rotatory power of an optically active solution is determined and found to be

i) By experiment $S = \text{----- deg/dm.}$

ii) By graph $S = \text{----- deg/dm.}$

TO FIND SPECIFIC ROTATION (S):

$$\text{TR} = \text{MSR} + (\text{VSR} \times \text{LC})$$

LC = 6'

Sl.No .	Concentration gm / cc	Initial Half shade			θ	θ / C
		MSR (deg)	VSR (div)	TR (deg)	(deg)	
1	Distilled Water					
2	0.1					
3	0.071					
4	0.056					
5	0.045					
6	0.038					

Mean

Viva Questions:

1. Define Polarization of light.

The Phenomenon of acquiring the property of one-sidedness in light is called polarization and the light is called polarized light.

2. What are polarized and Unpolarized light?

If the vibration of a light source is in all possible directions with respect to direction of propagation is called as unpolarised light.

When the vibration of the light are confined along a single direction of propagation the light is known as plane polarized.

3. Define Plane of vibration and plane of polarization

The plane of polarization is that plane in which no vibrations occur and the plane in which vibrations occur known as plane of vibration. The vibrations occur at the right angle to the plane of polarization.

4. Define optic axis of a crystal.

A line passing through any one of the blunt corners (A or B) and making equal angles with the three faces which meet at this corner, locate the direction of the optic axis.

5. Define double refraction.

When a light is passed through a calcite crystal, the single incident beam splits into two reflected rays

Ordinary ray (O-ray)

Extra- ordinary ray (E-ray)

6. Define Polarizer and Analyzer. Give examples.

Polariser: A device which converts an unpolarised light into a plane polarized light is called a polarizer.

Analyser: A device which check whether the light is plane polarized or not is called an analyzer

Example: Nicol Prism.

7. Distinguish between Ordinary ray and Extra-Ordinary ray.

Sl.No	Ordinary ray	Extra-ordinary ray
1	It obeys the laws of refraction	It does not obey the laws of refraction
2	It travels with a low velocity	It travels with a higher velocity than ordinary ray
3	Ordinary ray is stationary	Extra-ordinary ray revolves around the ordinary ray

8. What are Uniaxial and Bi-axial crystals?

A Uniaxial crystal is the one which has only one optic axis.

A Bi-axial crystal is the crystal which has two optic axis.

9. Give examples for positive and negative uniaxial crystals.

Positive uniaxial crystal: ice and quartz

Negative uniaxial crystal: Calcite and tourmaline

10. Define specific rotation.

The specific rotation is defined as the rotation produced by a decimeter (10cm) long column of the liquid containing 1 gram of the active substance in one cc of the solution. It is given by

$$S = \frac{100\theta}{LC}$$

11. Give examples for optically active solutions and crystals.

Sugar crystals, turpentine, sodium chlorate, cinnabar etc.,

12. What are dextro- rotatory and levo-rotatory substances?

The substances which rotate the plane of vibration to the right or clockwise direction are called dextro-rotatory or right handed rotation. The substances that rotate the plane of vibration to left or anticlockwise direction are called as laevo rotatory crystals.

13. Define Optical activity.

The phenomenon or the property of rotating the plane of vibration by certain crystals or substances is known as optical activity.

14. What is Brewster's law of polarization?

The tangent of the polarizing angle is mathematically equal to the refractive index of the material i.e $\mu = \tan (i_p)$

15. What do you mean by Snell's law?

The sine of the angle of incidence to sine of the angle of refraction is equal to the refractive index of the material.

LASER GRATING

Expt No:

Date:

AIM:

To determine the wavelength of given source of light using grating elements

APPARATUS REQUIRED:

Spectrometer, grating, screen, pencil, grating stand, semiconductor laser, etc.

FORMULA:

Wavelength of the laser, $\lambda = \frac{\sin \theta}{m N} \dots\dots\dots (\text{\AA})$

θ - Angle of diffraction (**deg**)

m – Order of diffraction

N – Number of lines in the grating (**Lines/m**)

PROCEDURE:

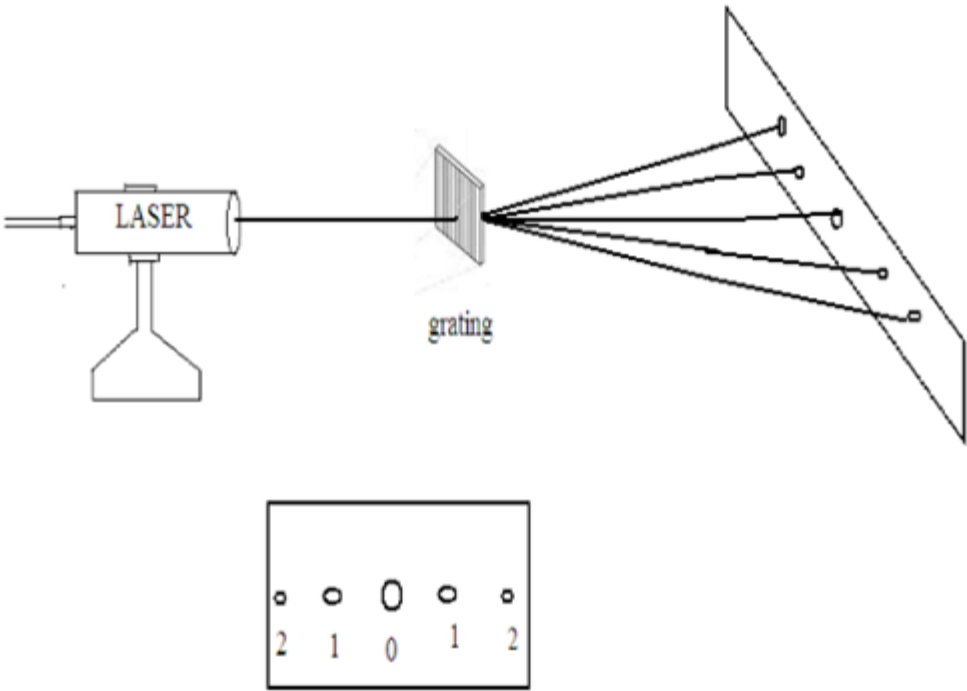
A Semiconductor laser is kept horizontally and switched on (extreme care should be taken to avoid direct exposure of laser light on eyes). The grating is held normal to the laser beam coincides with the beam coming out of the laser source.

After adjusting for normal incidence, the laser light is exposed to the grating and is diffracted by it. On the other side of the grating on the screen, the diffracted laser spots are seen. The distances of different orders from the center spot (X_n) are measured. The distance between the grating and the screen D is measured. Using the formula θ is calculated. The wavelength of the laser light is found with the formula $\lambda = \frac{\sin \theta}{m N}$

The number of lines in the grating is assumed to be as 15,000 lines per inch, it may be taken as 59285 lines per meter. It can also be measured from the experiment using known wavelength (sodium vapour lamp) of the standard source and using the

formula $N = \left(\frac{\sin \theta}{m \lambda} \right) \text{lines / m}$

Diagram:



To find the Number of Lines in the Grating: $TR = MSR + (VSC \times LC)$ $LC = 1'$

Diffracted Ray	Telescope Reading			Telescope Reading			Difference (deg)		Mean $\theta = \left(\frac{V_A + V_B}{4} \right)$	$N = \frac{\sin \theta}{m \lambda}$
	Vernier A			Vernier B			V _A	V _B		
	MSR (deg)	VSC (div)	TR (deg)	MSR (deg)	VSC (div)	TR (deg)	2θ	2θ		
LEFT										
RIGHT										

To find wavelength of the laser source:

[illegible]

Determination of Least Count of a Spectrometer:

$$\text{Least Count} = (1\text{MSD} - 1\text{VSD})$$

$$2 \text{ MSD} = 1^\circ$$

$$\therefore 1 \text{ MSD} = \left(\frac{1}{2}\right)^\circ$$

$$30 \text{ VSD} = 29 \text{ MSD}$$

$$1 \text{ VSD} = \left(\frac{29}{30}\right) \text{ MSD}$$

$$1 \text{ VSD} = \left(\frac{29}{30}\right) \times \left(\frac{1}{2}\right)^\circ$$

$$1\text{VSD} = \left(\frac{29}{60}\right)^\circ$$

$$\text{Least count} = \left(\frac{1}{2}\right)^\circ - \left(\frac{29}{60}\right)^\circ$$

$$\text{Least count} = \left(\frac{30 - 29}{60}\right)^\circ = \left(\frac{1}{60}\right)^\circ = 1'$$

$$\therefore \text{Least count} = 1'$$

RESULT:

Wavelength of the laser light source = ----- Å

Viva Questions:

1. Define LASER.

Light amplification by stimulated emission of radiation.

2. Define Population Inversion.

When the no. of atoms are more in higher energy state than the lower energy state, is known as population inversion. it is essential for stimulated emission

3. List out the properties of LASER.

Directionality, Monochromaticity, High intensity, Coherence.

4. What will happen if light radiation interacts with matter?

If light interacts with matter (group of atoms or molecules) the light is absorbed or emitted by particles (atoms or molecules)

5. Define Stimulated emission and Spontaneous emission.

(i) Spontaneous emission:

The atom in the excited state returns to the ground state by emitting a photon of energy $E = E_2 - E_1 = h\nu$, spontaneously without any external triggering as shown in fig. This process is known as spontaneous emission.

(ii) Stimulated emission:

The atom in the excited state can also return to the ground state by external triggering (or) inducement of photon thereby emitting a photon of energy equal to the energy of the incident photon, known as stimulated emission.

6. Differentiate between ordinary light and laser light.

S.No	Ordinary light	Laser light
1	In ordinary light the angular spread is more.	In laser beam the angular spread is less.
2	They are not directional.	They are highly directional.
3	It is less intense	It is highly intense.
4	It is not a coherent beam and is not in phase.	It is a coherent beam and is in phase.
5	The radiations are polychromatic.	The radiations are monochromatic

7. How will you achieve population inversion in laser?

The excitation of atoms in order to achieve the population inversion can be done by the help external energy is referred as pumping..

8. List out the different types of pumping mechanism to achieve population inversion.

Types of pumping are Optical pumping, direct electron pumping, Inelastic atom-atom collisions, Chemical reaction

9. List out the applications of laser in various fields.

Laser beams are used to obtain a three dimensional lens less holograms.

Laser is used in printers for computer printouts.

It is used in transmitting signals to longer distances.

It is used in long haul communication system.

It is also used in the endoscopic applications.

It plays a vital role in remote sensing applications.

It is used to carry out blood less micro surgeries.

10. What is diffraction?

When the light bends or deviates from path due to obstacle known as diffraction.

11. Define Grating

It is an arrangement in which there is a fine quality of glass on which parallel lines are ruled equidistant with diamond point by an automatic shifting machine. The ruled lines works as opaque and spacing works as transparent.

12. What is active medium?

It may be solid, liquid, or gas which achieves population inversion.

13. What is the action of the optical resonator?

It is the combination of two reflecting mirror which is used to increase the intensity of laser light.

14. What are the different types of laser?

Based on the active medium, they are classified as Solid state lasers, Gas lasers, Liquid Lasers, Dye lasers and Semiconductor lasers.

15. What are the conditions needed for laser action?

1. The rate of absorption should be equal to rate of emission.
2. Stimulated emission should be prominent over spontaneous emission and absorption.
3. Number atoms in the excited state must be higher than the ground state.
4. Losses in the lasing medium should be overcome by induced transition in the amplifying medium.

THERMAL CONDUCTIVITY OF RUBBER BY RADIAL FLOW METHOD

Expt No:

Date:

AIM:

To determine the thermal conductivity of rubber using by radial flow method.

APPARATUS REQUIRED:

Rubber tube, steam boiler, calorimeter, stirrer, thermometer, stop clock, meter scale.

FORMULA:

$$K = \left[\frac{\{W_1 C_1 + (W_2 - W_1) C_2\} (\theta_2 - \theta_1)}{2 \pi l t} \right] \left[\frac{2.303 \log_{10} \left(\frac{r_2}{r_1} \right)}{\theta_s - \left(\frac{\theta_1 + \theta_2}{2} \right)} \right] \dots\dots \mathbf{W/m/K}$$

K – Thermal Conductivity of the given Specimen

W₁ – Weight of empty calorimeter & stirrer (**Kg**)

W₂ – Weight of calorimeter, stirrer & water (**Kg**)

C₁ – Specific heat capacity of calorimeter = 385 (**J/Kg/K**)

C₂ – Specific heat capacity of water = 4190 (**J/Kg/K**)

l – Length of the rubber tube immersed in water (**m**)

r₁ – Inner radius of the tube (**m**)

r₂ – Outer radius of the tube (**m**)

θ_s – Steam temperature (°C)

θ₁ – Initial temperature (°C)

θ₂ – Final temperature (°C)

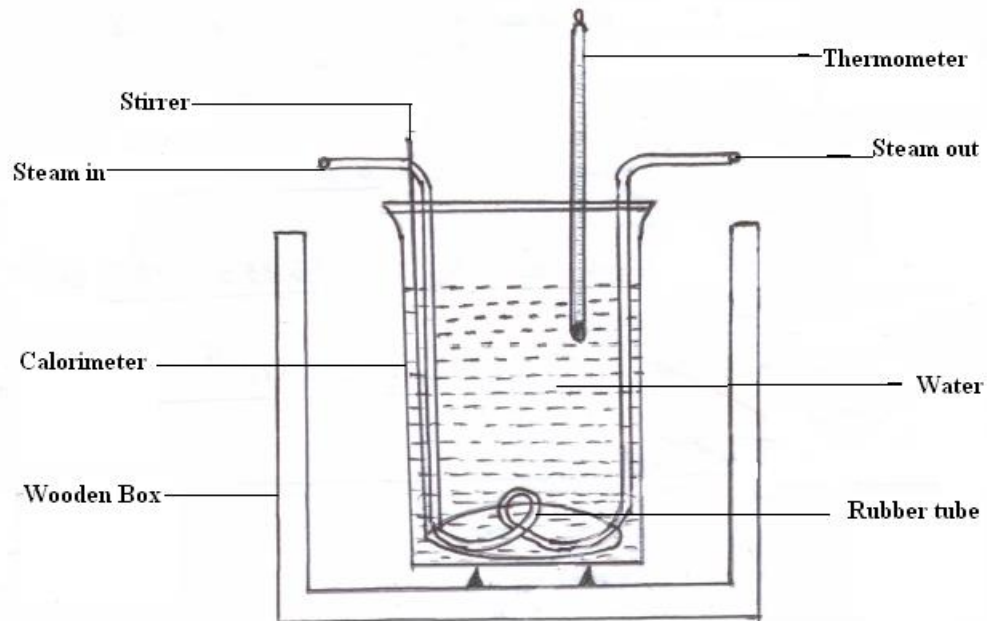
t – Time taken for temperature rise temperature from initial to final (s)

PROCEDURE:

A large calorimeter of capacity 500 ml is weighed empty with stirrer (W₁). It is filled two-thirds with cold water in room temperature and again weighed (W₂). A plentiful supply of steam is produced by heating the water in the boiler. The calorimeter is placed inside a wooden box packed with non-conducting cotton wool. By means of a wooden screen, the calorimeter is protected, from the direct from the heater.

A known length (l) of a rubber tube is cooled, and immersed in water contained in the calorimeter with both ends of the tube projecting some distance outside the calorimeter. The length of the rubber tubing kept immersed inside the water is indicated by two fine threads tied around the tube, were it just emerges out of water.

Diagram:



Thermal Conductivity of Rubber

To find rate of rise of temperature:

1	Initial temperature °C	Time (sec)
2	Final temperature °C	Time (sec)

To find the inner and outer radius of the rubber tube:

$$\text{TR} = \text{MSR} + (\text{VSC} \times \text{LC})$$

$$\text{LC} = 0.001\text{cm}$$

Sl.NO	Position of the Microscope	Microscope Reading						Diameter X 10 ⁻² m	Radius X10 ⁻² m
		Outer			Inner				
		MSR cm	VSC div	TR cm	MSR cm	VSC div	TR cm		
1.	Left							(d ₂ - Outer)	(r ₂ - Outer)
2.	Right							(d ₁ - Inner)	(r ₁ - Inner)

LEAST COUNT DERIVATION FOR TRAVELLING MICROSCOPE:

$$\text{Least count} = 1\text{MSD} - 1\text{VSD}$$

$$20\text{MSD} = 1\text{cm}$$

$$\text{Value of 1MSD} = \left(\frac{1}{20}\right)\text{cm}$$

$$\text{Number of vernier scale division VSD} = 50$$

$$\therefore 50\text{VSD} = 49\text{MSD}$$

$$\therefore 1\text{VSD} = \left(\frac{49}{50}\right)\text{MSD} = \left(\frac{49}{50}\right) \times \left(\frac{1}{20}\right)\text{cm} = \left(\frac{49}{1000}\right)\text{cm}$$

$$\text{LC} = \left(\frac{1}{20}\right)\text{cm} - \left(\frac{49}{1000}\right)\text{cm}$$

$$= \left(\frac{50 - 49}{1000}\right)\text{cm} = \left(\frac{1}{1000}\right)\text{cm}$$

$$\therefore \text{LC} = 0.001\text{cm}$$

The calorimeter is stirred well and the initial temperature $\theta_1^\circ\text{C}$ is noted. Now one end of the rubber tube is connected to steam generator, and steam is passed through it, at once starting a stop clock. The calorimeter is kept stirred well and the temperature is noted with the thermometer kept immersed in it. The steam is passed continuously till there is a rise of 10°C in temperature. The time (t seconds) taken for this rise in temperature is noted.

Let $\theta_2^\circ\text{C}$ be the corrected final temperature of the calorimeter. The internal radius (r_1) and the external radius (r_2) can be determined by taking an ink-print of the cross section of the tube, and measuring with a vernier microscope. From the readings the thermal conductivity of the rubber is determined.

RESULT:

The thermal Conductivity of the rubber tube is determined by radial flow method $K = \text{----- w/m/k.}$

Viva Questions:

1. Differentiate Conduction, Convection and Radiation.

Conduction: It is the process in which heat is transferred from region of higher temperature to the region of lower temperature of body without any transfer of matter or molecules

Convection: It is the process in which particles of the material at higher temperature move to cold region and particles at low temperature move to hotter region setting up a cyclic flow of material.

Radiation: It is the process in which heat is transferred by electromagnetic waves of wavelength from about 7500 Å to about 3mm. these waves do not need any material medium.

2. Classify materials based on the property of conduction.

In non-metallic solids, liquids, gases, such collisions being few, heat transfer is slow process, hence these are bad conductor.

In metals, free electrons are present which move rapidly and collide very frequently, thus transferring heat. Due to free electrons, metals are good conductors of electricity.

3. Differentiate between Conductor, Semiconductor and Insulator.

Conductor

1. The conductor is the material which has low resistivity
2. The conduction band and valence band overlaps and there is no forbidden band gap.

Semiconductor

1. The semiconductor is the material which has resistivity lying between a conductor and an insulator
2. The conduction band and valence band is separated by a forbidden band gap of energy Eg. Insulator.

1. Insulator is the material which has zero resistivity
2. The conduction band and valence band is separated by a large distance forbidden band gap of energy Eg.

4. Why conductors are good in conducting heat and electricity.

1. The electrical and thermal conductivity decreases with the increase in temperature and impurities
2. The electrical and thermal conductivity is very high at low temperature
3. For non-metal, the electrical and thermal conductivity is very less

5. Define specific heat capacity.

The quantity of heat required to raise the temperature of unit mass of the substance through 1 kelvin.

$$S = \frac{Q}{m\theta} \text{ J/kg/K}$$

6. Define Co-efficient of thermal conductivity.

The quantity of heat conducted per unit area per unit time maintained at unit temperature gradient

7. Define temperature gradient.

The rate of fall of temperature with respect to distance is called temperature gradient.

The smaller values $\frac{\theta_1 - \theta_2}{x}$ is written as $\frac{d\theta}{dx}$. It is known as the temperature gradient.

8. What is meant by radial flow of heat ?

